

Data Visualisation

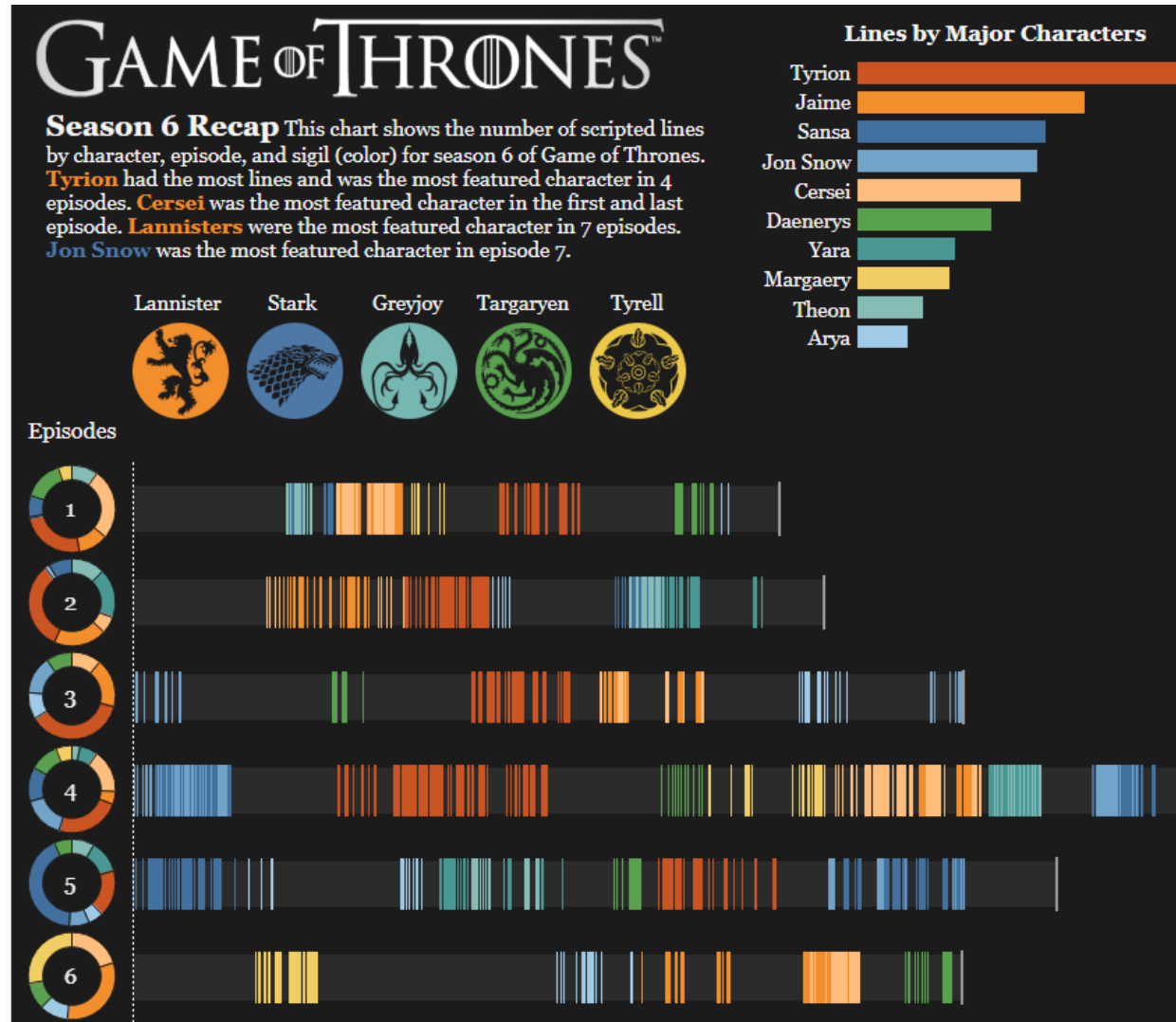
Waiting for students to log in

Data Visualisation

Lecture 2

Dr. Marisa Llorens-Salvador

Visualisation of the week



UnVisualisation of the week



Two views:

1. Tufte advocates a minimalist perspective (High data to ink ratio).
2. Improved recall when using visual embellishments.

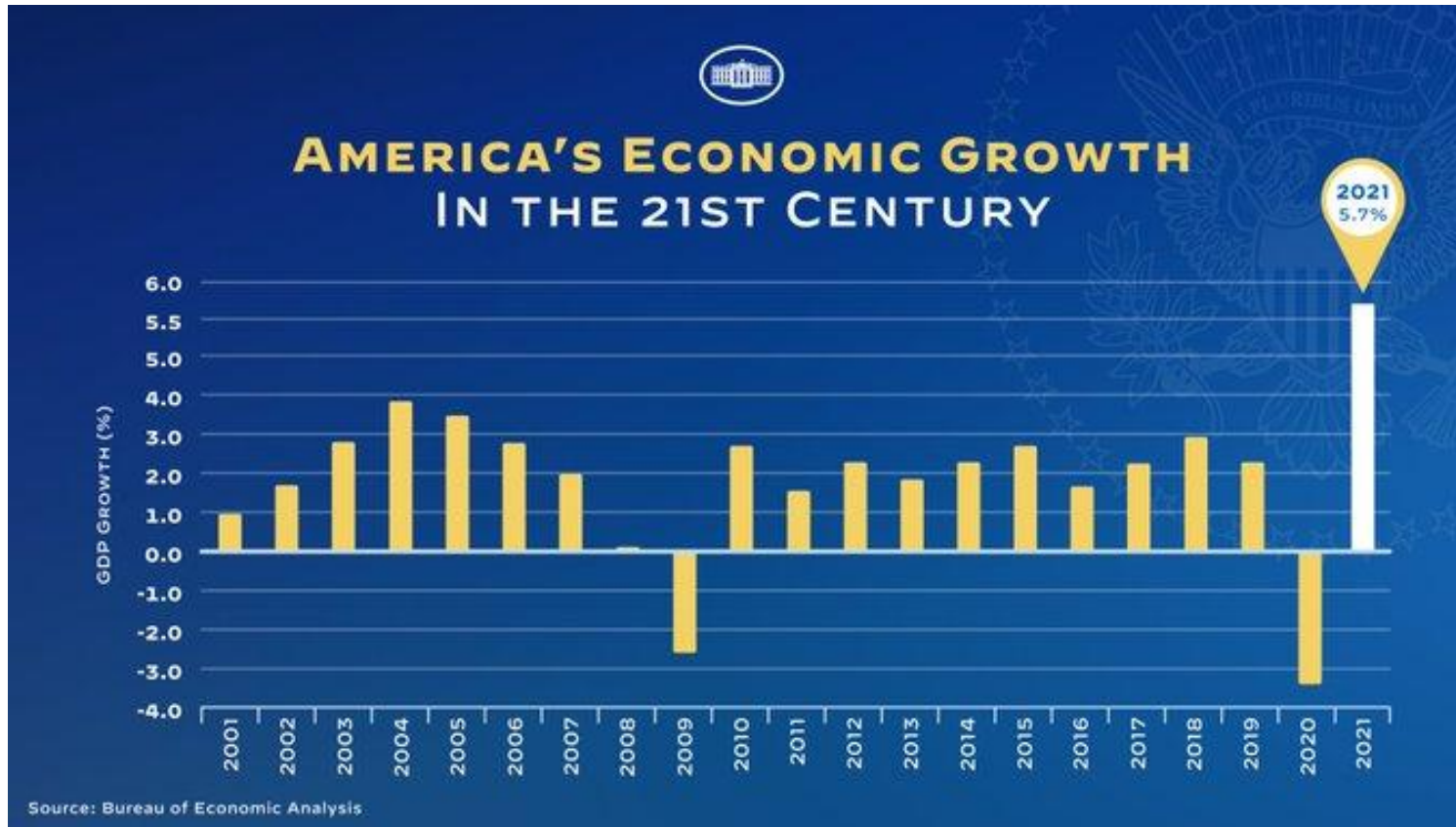
Useful Junk? The Effects of Visual Embellishment on Comprehension and Memorability of Charts

Scott Bateman et al. Published in *In the proceedings of Proceedings of the SIGCHI conference on human factors in computing systems*, 2010

Discussion

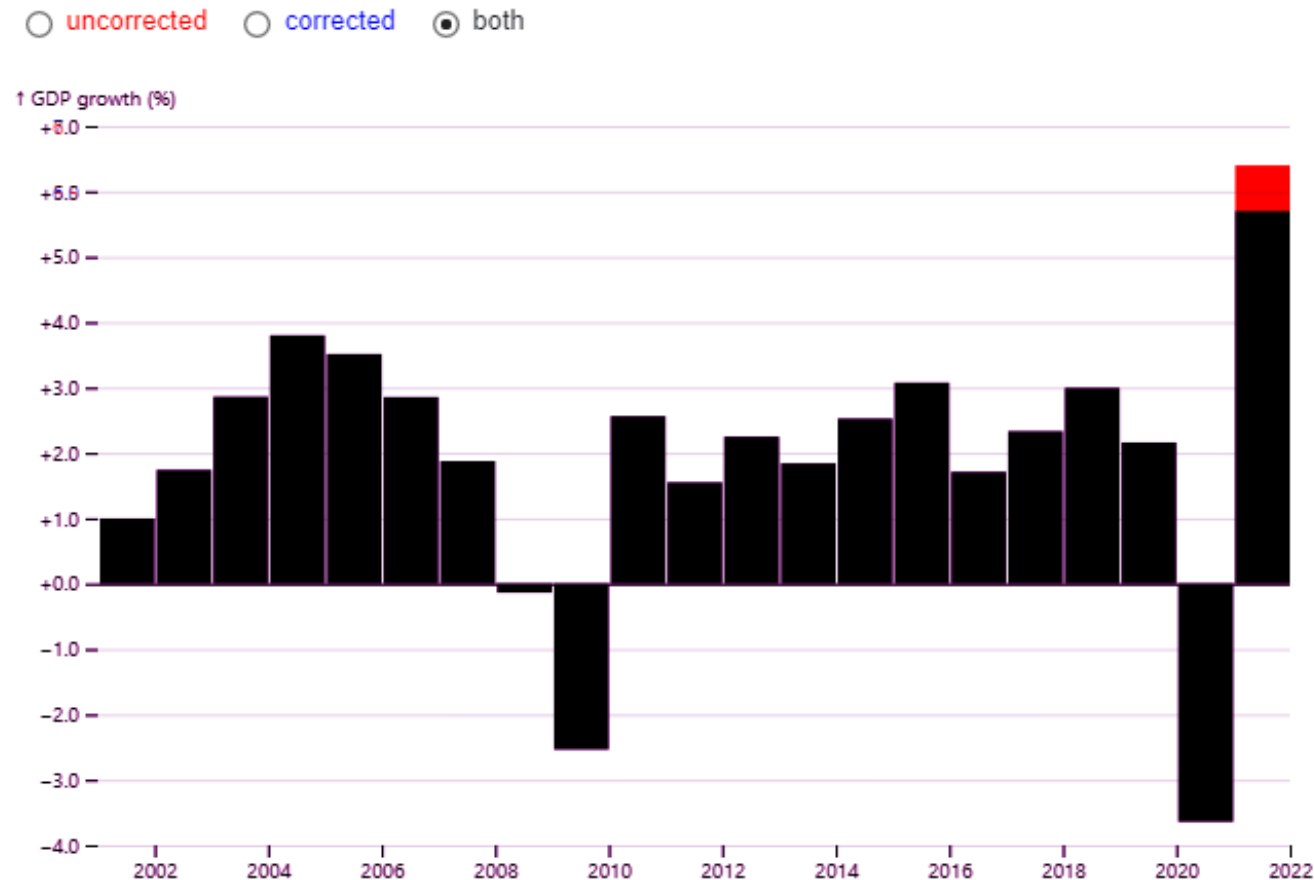
The White House Tweet 27th Jan 2022

We just learned that President Biden's first year in office was the strongest year for economic growth since 1984.



Discussion

The White House Tweet 27th Jan 2022



<https://observablehq.com/@mbostock/a-misleading-plot>

Remember!

- *Well-designed visual representations can replace cognitive calculations with **simple perceptual inferences** and **improve comprehension, memory, and decision making***

Human memory and Visualisation

- The power of visualisation is the ability to get across lot more complex information than what our memory can generally hold
- There are 3 types of memories that process information in our brain:
 1. Iconic memory or Sensory memory
 2. Working memory
 3. Long term memory

Human Memory and Visualisation

Type of memory	Iconic memory	Working memory	Long term memory
Duration	1 second approx.	1 minute approx.	1 second -lifetime
Processing	Pre-attentive processing, before we pay attention	Hold and process about 5-9 chunks of information (Miller's Law)	Information stored through rehearsal or repeated application

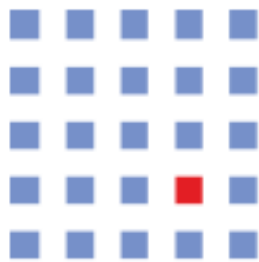
- Long term memory is where things we memorize or remember are stored
- Iconic and working memories are the ones that interact with visualisations, so let's look at them in depth

Iconic Memory

- When we see a visual, the information remains in the iconic memory for less than a second. We process and store information automatically in this time
- This process is called **pre-attentive processing** and it happens automatically, before we pay attention to the information

Iconic Memory

- The pre-attentive process detects several visual attributes
- Understanding how to make a particular attribute stand out can help us create visuals that emphasize the more important information



Colour



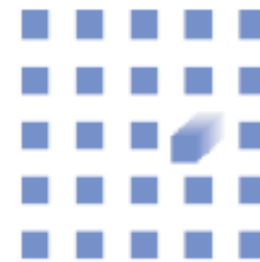
Orientation



Shape

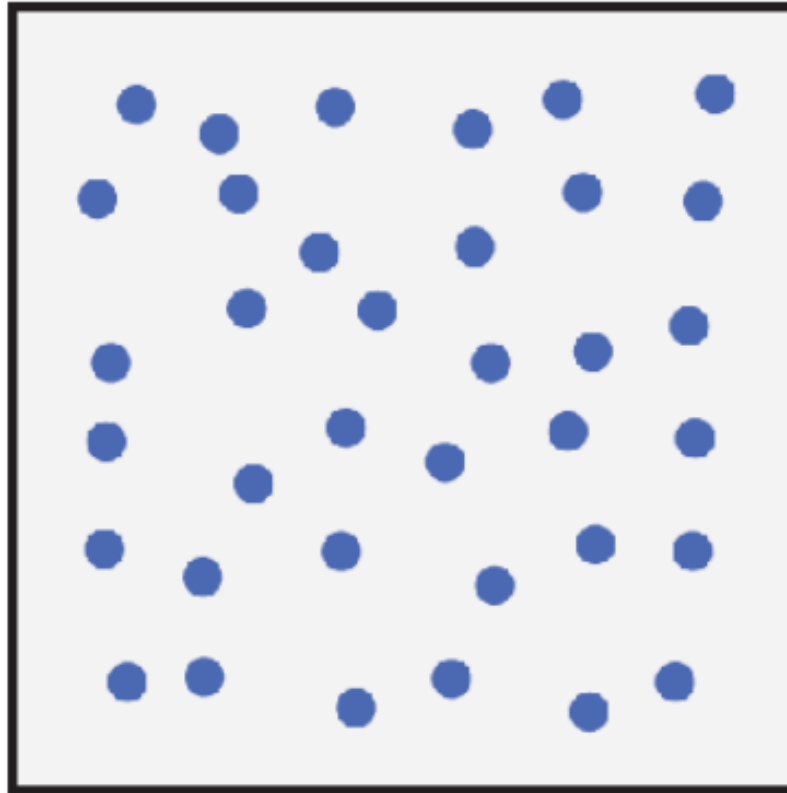


Brightness

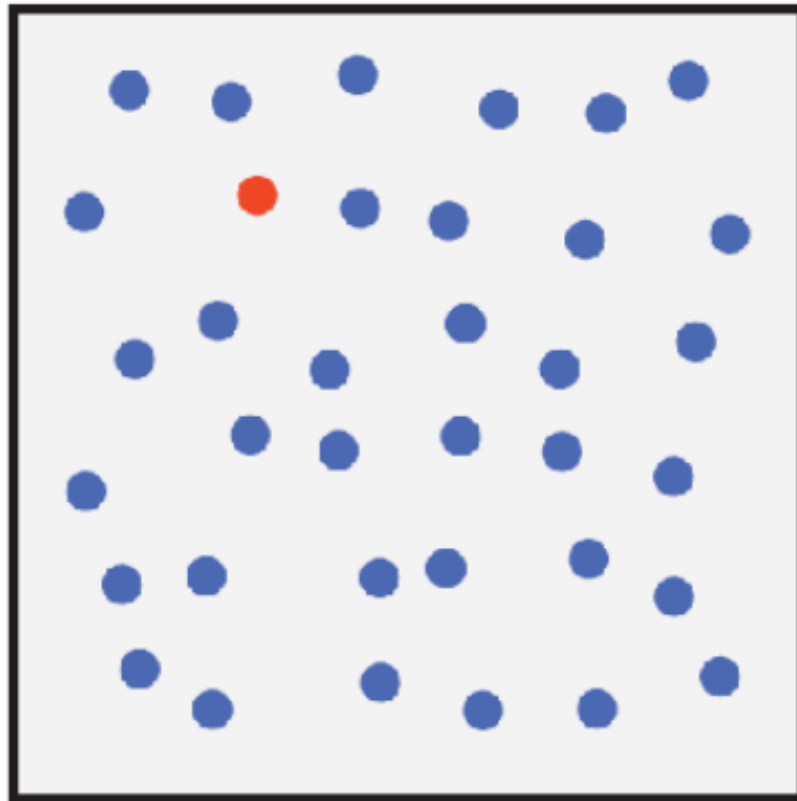


Movement

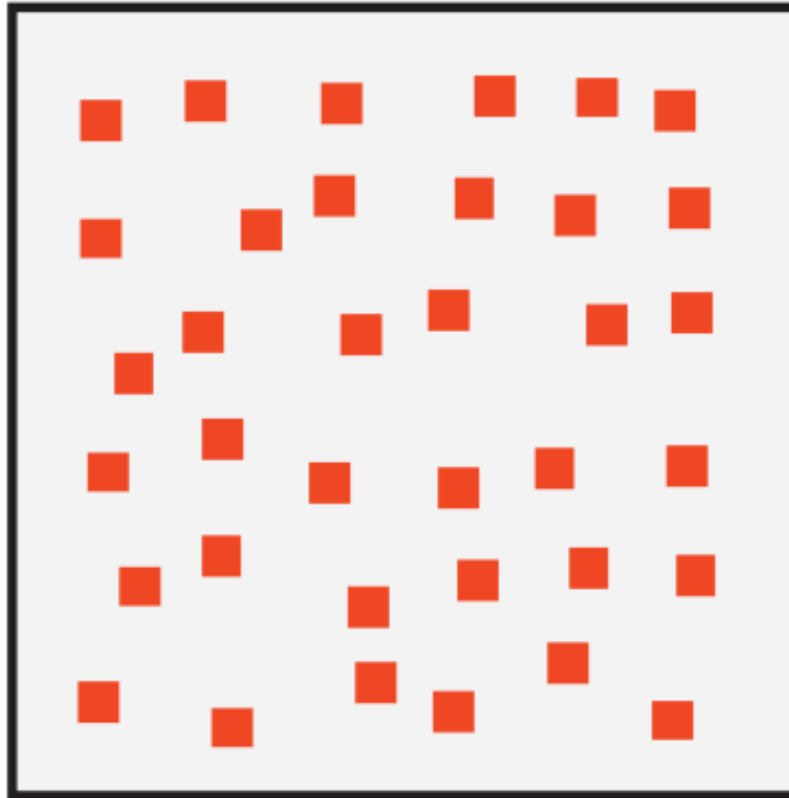
Is there a red circle?



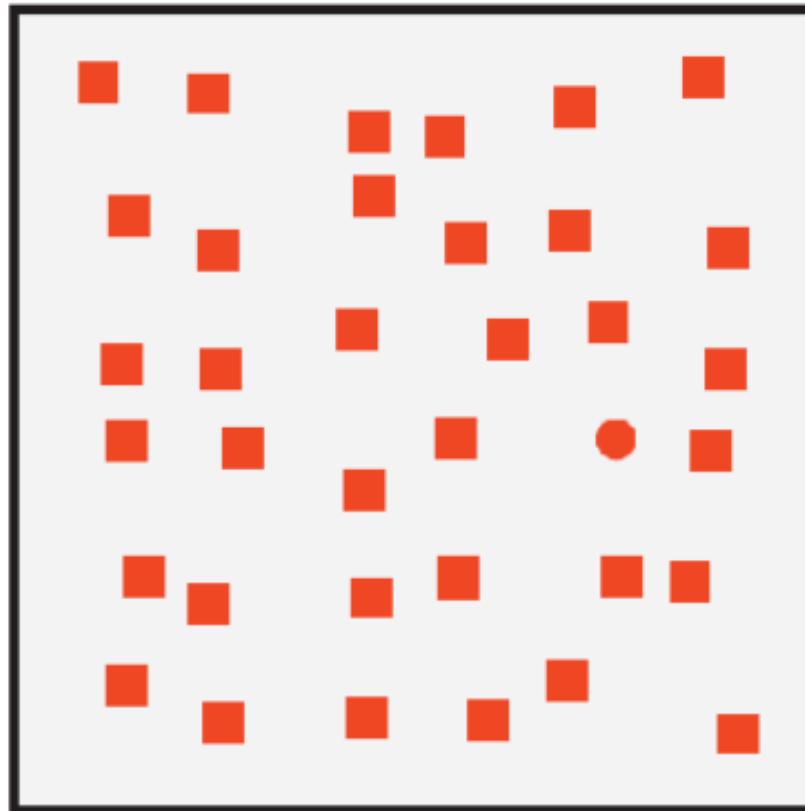
Is there a red circle?



Is there a red circle?



Is there a red circle?



How many 5's can you find?

987349790275647902894728624092406037070570279072
803208029007302501270237008374082078720272007083
247802602703793775709707377970667462097094702780
927979709723097230979592750927279798734972608027

How many 5's can you find?

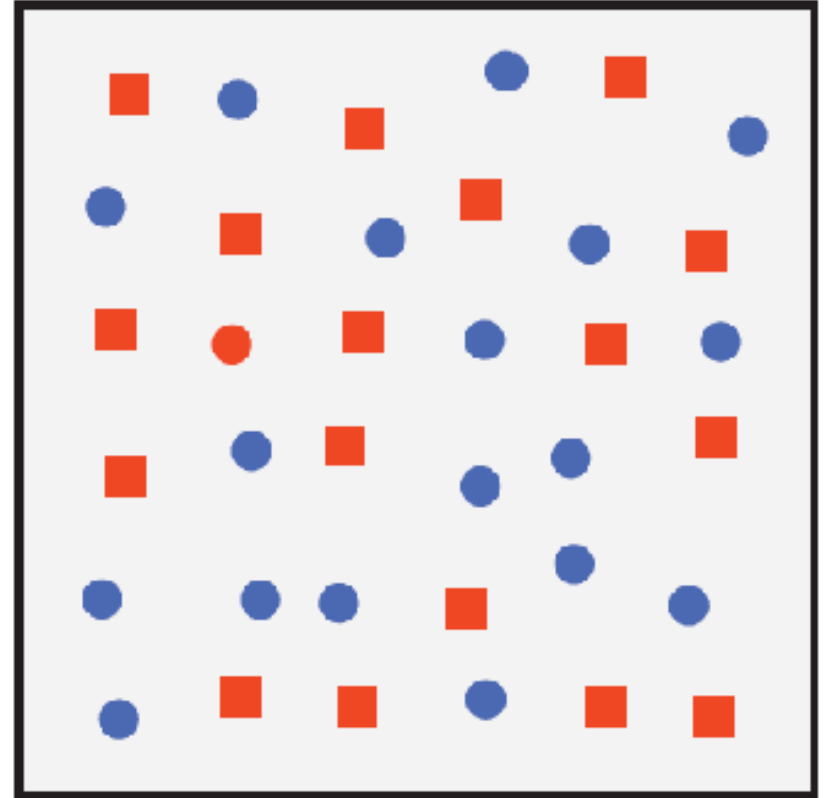
98734979027**5**647902894728624092406037070**5**70279072
803208029007302**5**01270237008374082078720272007083
24780260270379377**5**709707377970667462097094702780
927979709723097230979**5**927**5**0927279798734972608027

Conjunction Targets

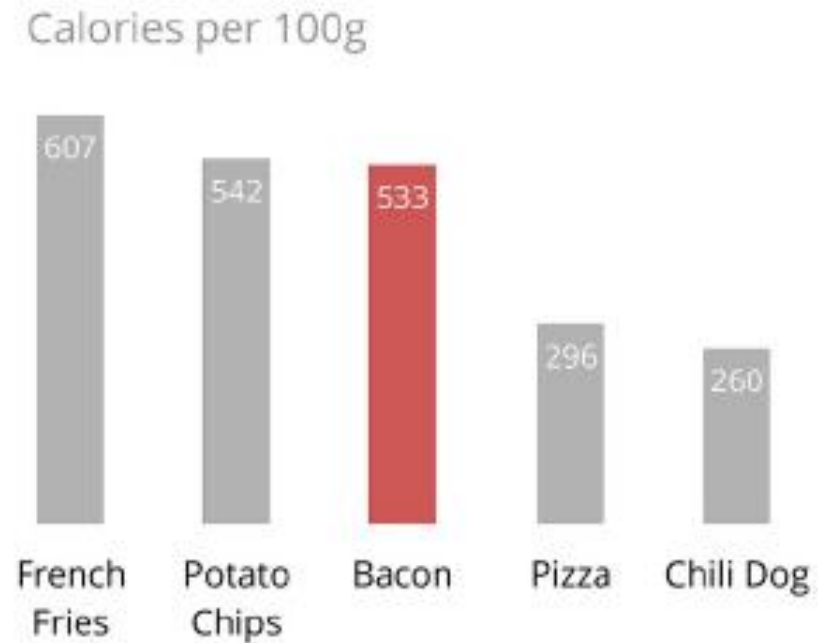
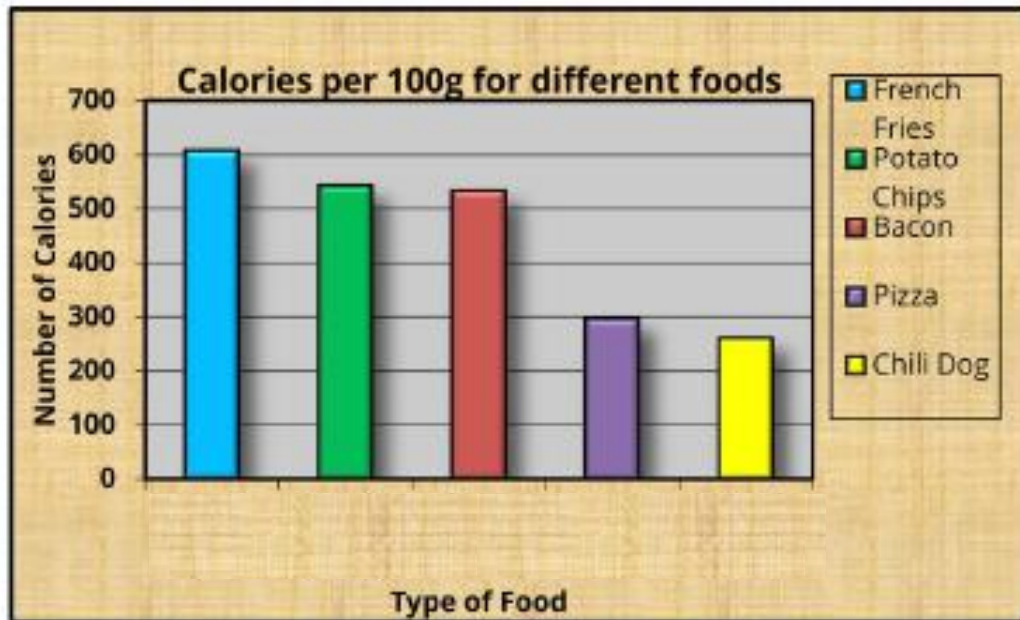
Two or more visual properties present which renders a target difficult to find

- Numerous studies have shown that most conjunction targets cannot be detected **pre-attentively**
- Viewers must perform a time consuming serial search through the display to confirm presence or absence

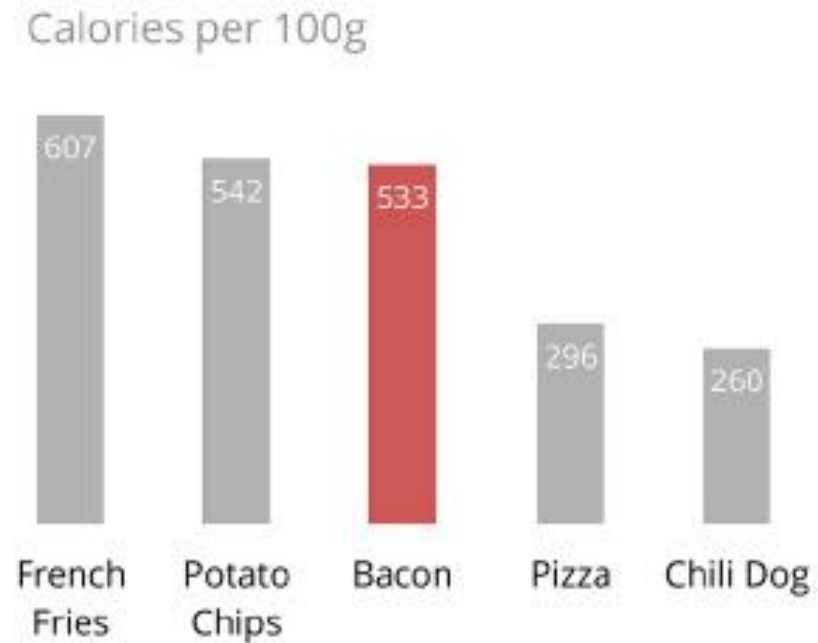
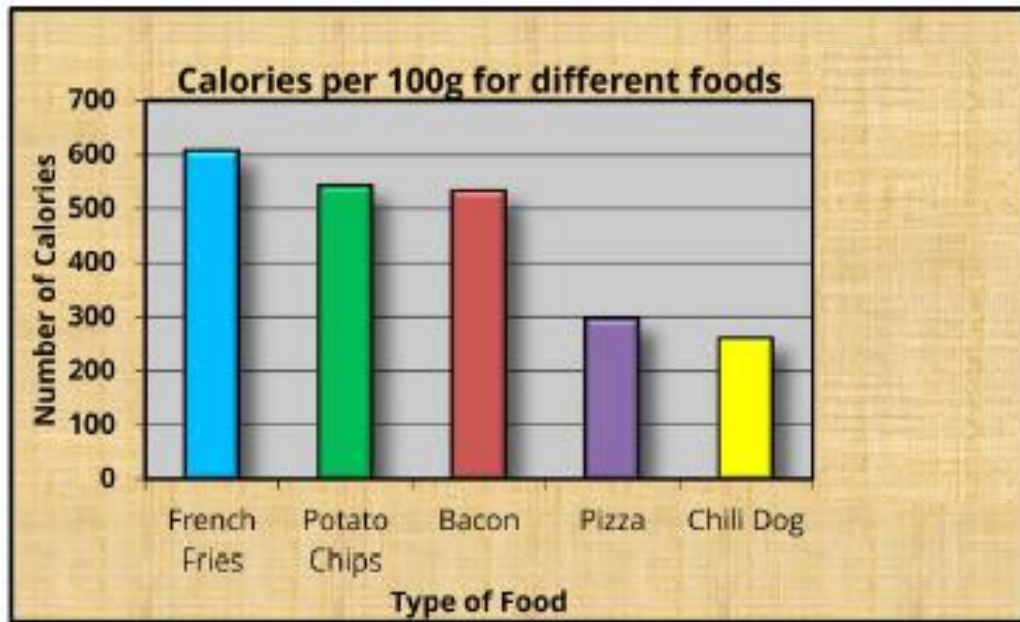
Is there a red circle?



Application to visualisation example



Application to visualisation example



Working Memory

- The memory in use when working with a visual
 - Sensory information of interest to us is processed in the working memory
 - Information stays here for about a minute
 - Capacity of our working memory is between 5 to 9 similar items (Miller's Law)
- The capacity of our working memory can be increased by a process called **chunking**
 - Grouping similar items together

Human Perception for Visualisation

- In order for our visualisations to be useful, it is important that they are easily interpreted by a viewer. Bad data visualisations can be difficult to read and understand
- To visualize data effectively, it is useful to follow design principles
 - Derived from an understanding of human perception
- In order to create effective visualisations we must understand perception

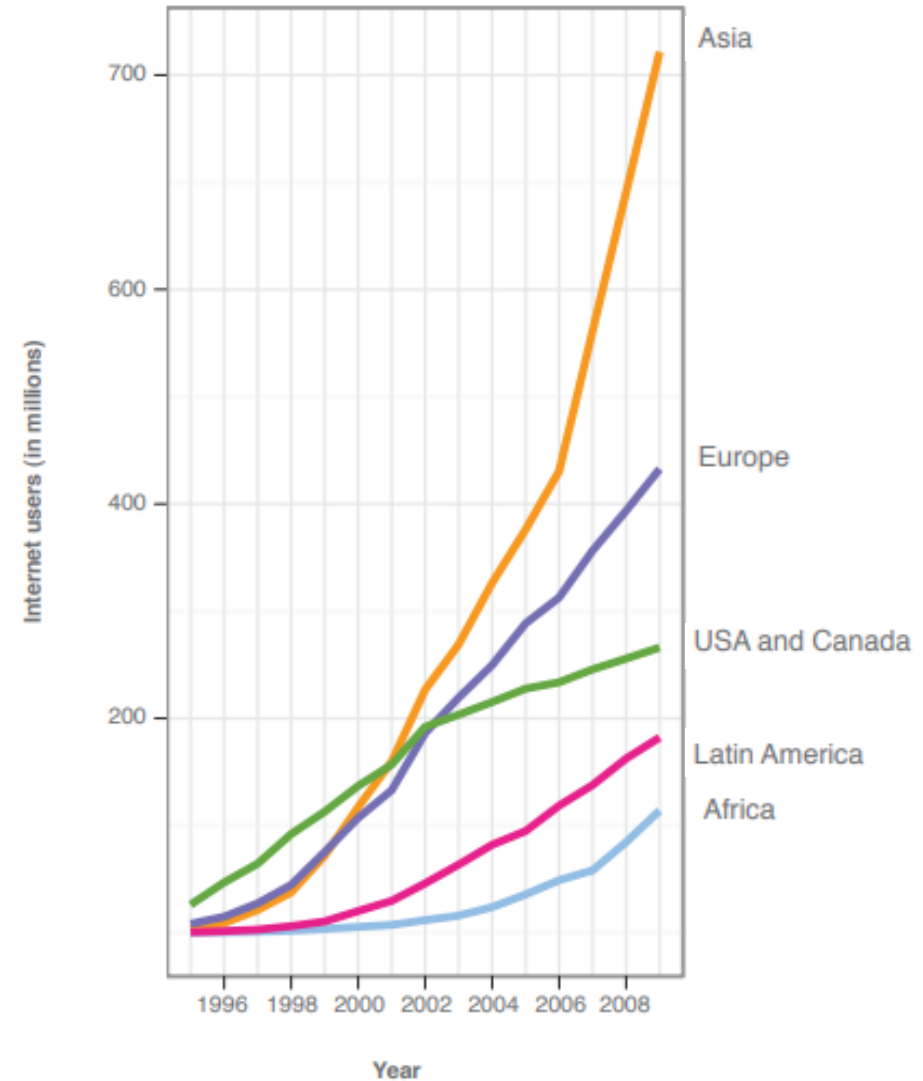
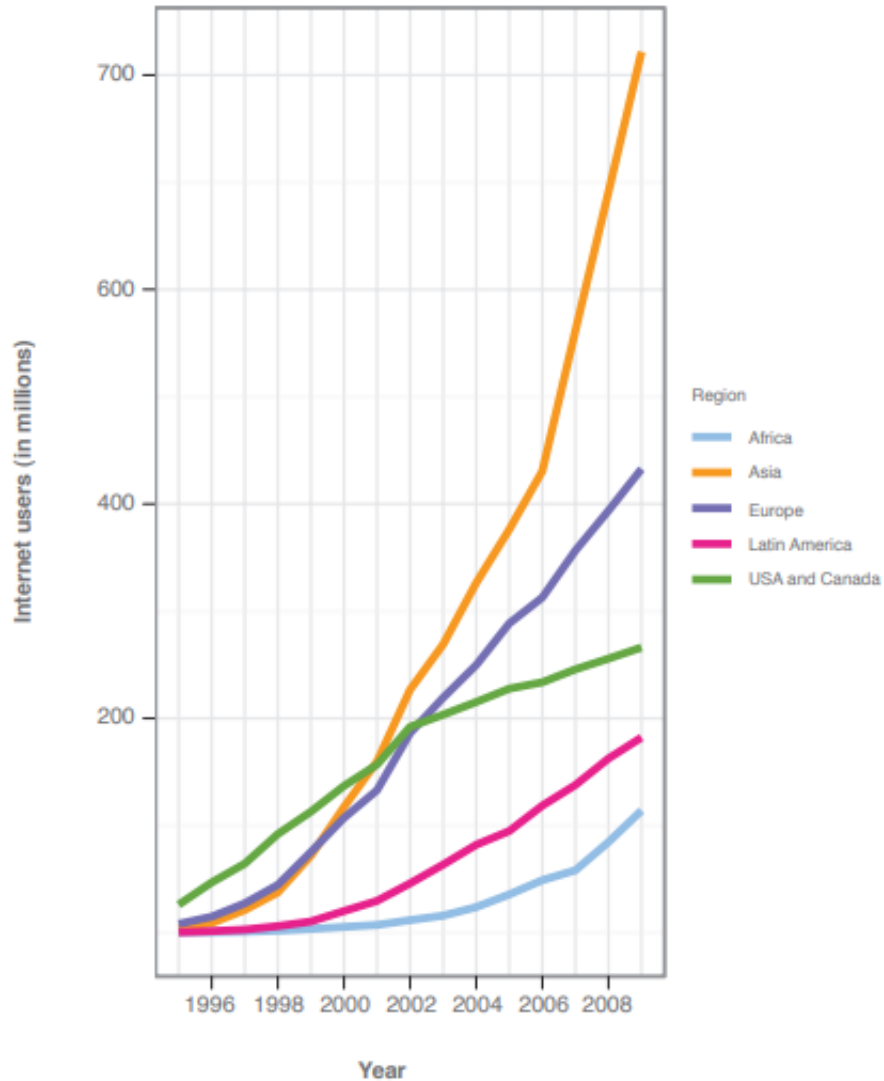
What is Human Visual Perception

- Our vision system has parallelism capacity
 - while it focuses on one point, the visual system can reach a wide surrounding area
- The human sensorial system is stimulated by a continuous flow of events
 - Resulting in a neural excitement called **sensation**
 - This process is extremely **fast and efficient**
 - This continuous flow of sensations generates what it is known as **perception**
- Perceptions are different according to the characteristics of the stimulus
 - They are **interpreted**, allowing the brain to extract knowledge
 - The act of thinking, processing and making comparisons is known as **cognition** and is **much slower and less efficient**

Human Visual Perception and Data Visualisation

- Data visualisation is an external aid to support our working memory
- Visual perception is selective
 - If we tune our awareness to everything, we will be overwhelmed. So we selectively pay attention to things that catch our attention
- Our eyes are drawn to familiar patterns
 - We see what we expect to see. So visualisation must take into account what people know and expect
- Our working memory is very limited
 - We can hold a very limited amount of information in our memory when looking at a visual
- We want more visual perception and less cognition

Two versions of the same visualisation



Gestalt Theory

- *“It is not possible to have knowledge of the whole through the parts, but of the parts through the whole; the whole is more than the sum of its parts”*
- Through the perception of the totality, the brain can realize, decode and assimilate an image or a concept
- It proposes a number of laws that explain the structural and functional principles of the perceptive

Some Laws of Gestalt Theory

- Proximity: Objects that are close together are perceived as a group
- Similarity: Objects that share similar attributes are perceived as a group
- Enclosure: Objects that appear to have a boundary around them are perceived as a group
- Closure: Open structures are perceived as closed, complete, and regular whenever there is a way that they can be reasonably interpreted as such
- Continuity: Objects that are aligned together or appear to be a continuation of one another are perceived as a group
- Connection: Objects that are connected are perceived as a group



Classifications of Visualizations

1

Complexity

2

Infographics

Data Vis

3

Exploration

Explanation

4

Informative

Persuasive

Visual Art

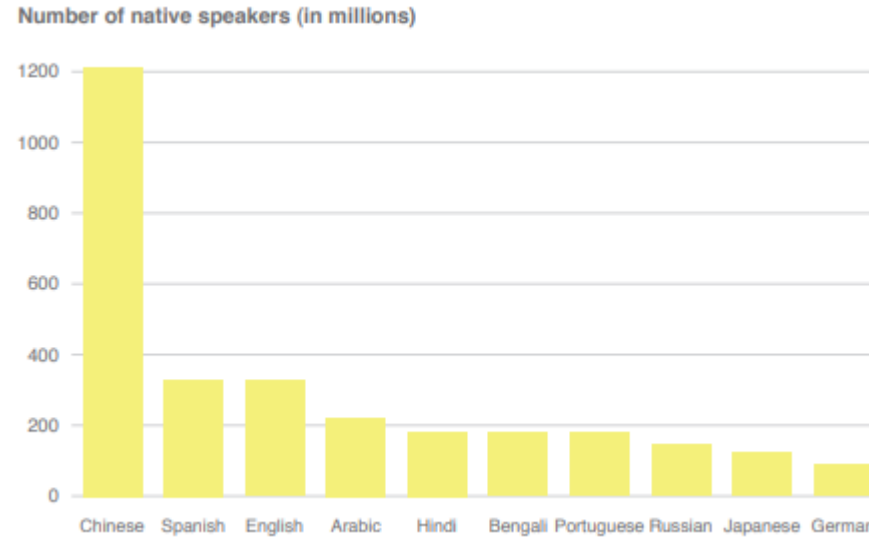
Complexity

Classify by counting how many different data dimensions are represented

- The number of discrete types of information that are visually encoded in a diagram

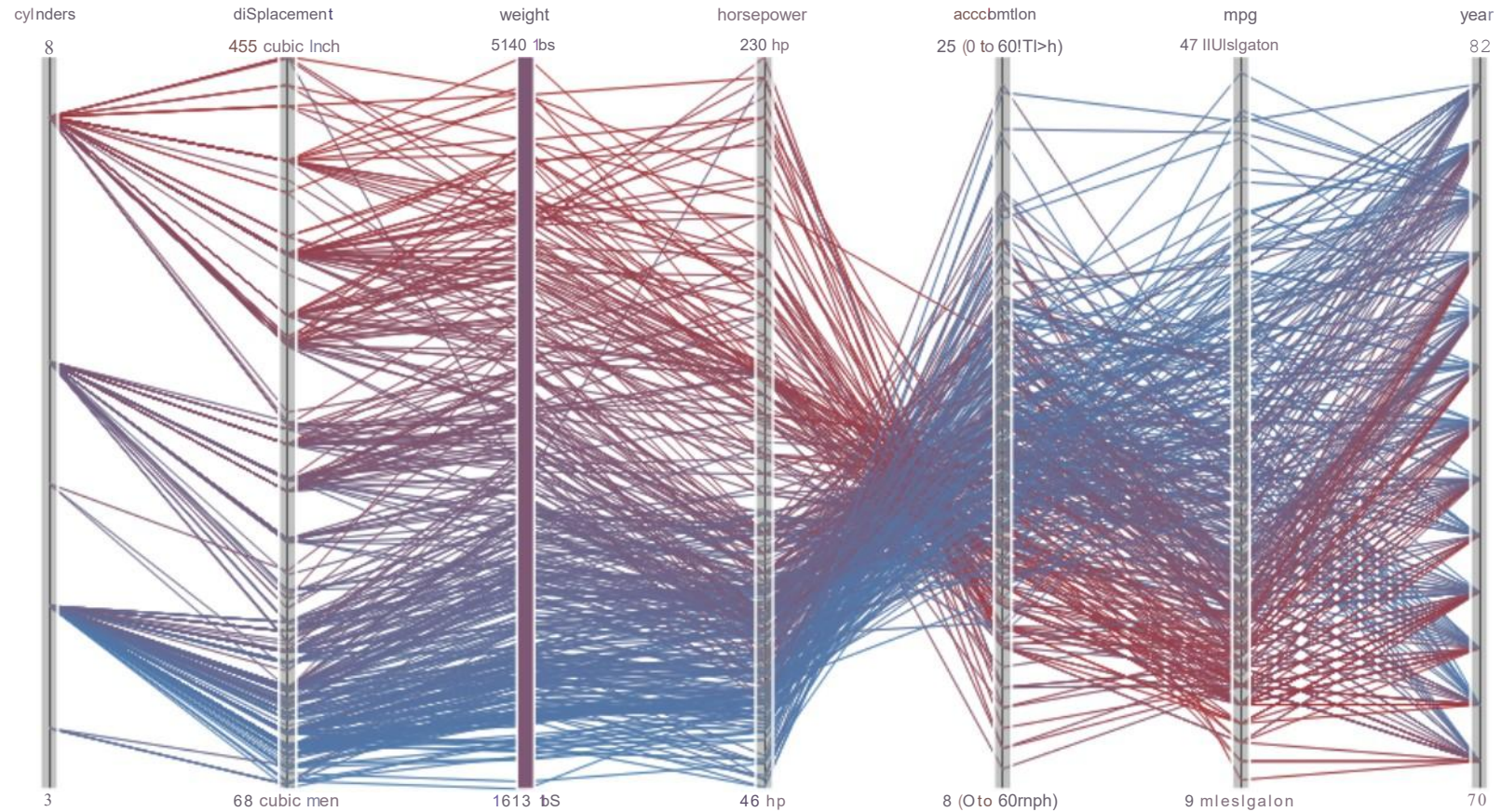
Complexity

- For example, a simple bar graph may show the number of native speakers per language: **2 data dimensions**



Complexity

Parallel Coordinates of Automobile Data

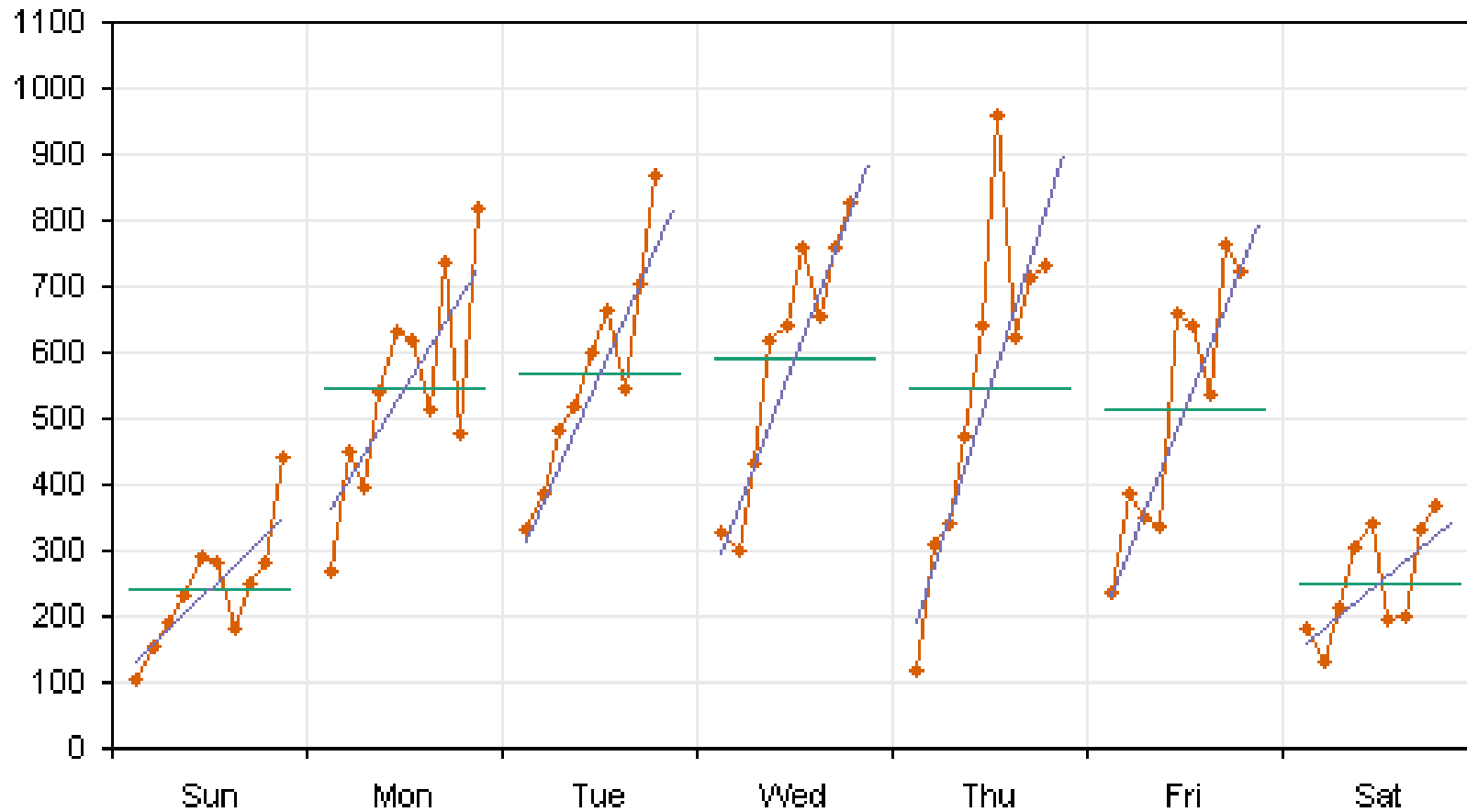


A database of cars is plotted in seven coordinate dimensions; each path represents one car. Drag and resize the coordinate selection sliders to filter the cars in any dimension.

Source: GGobi

Complexity

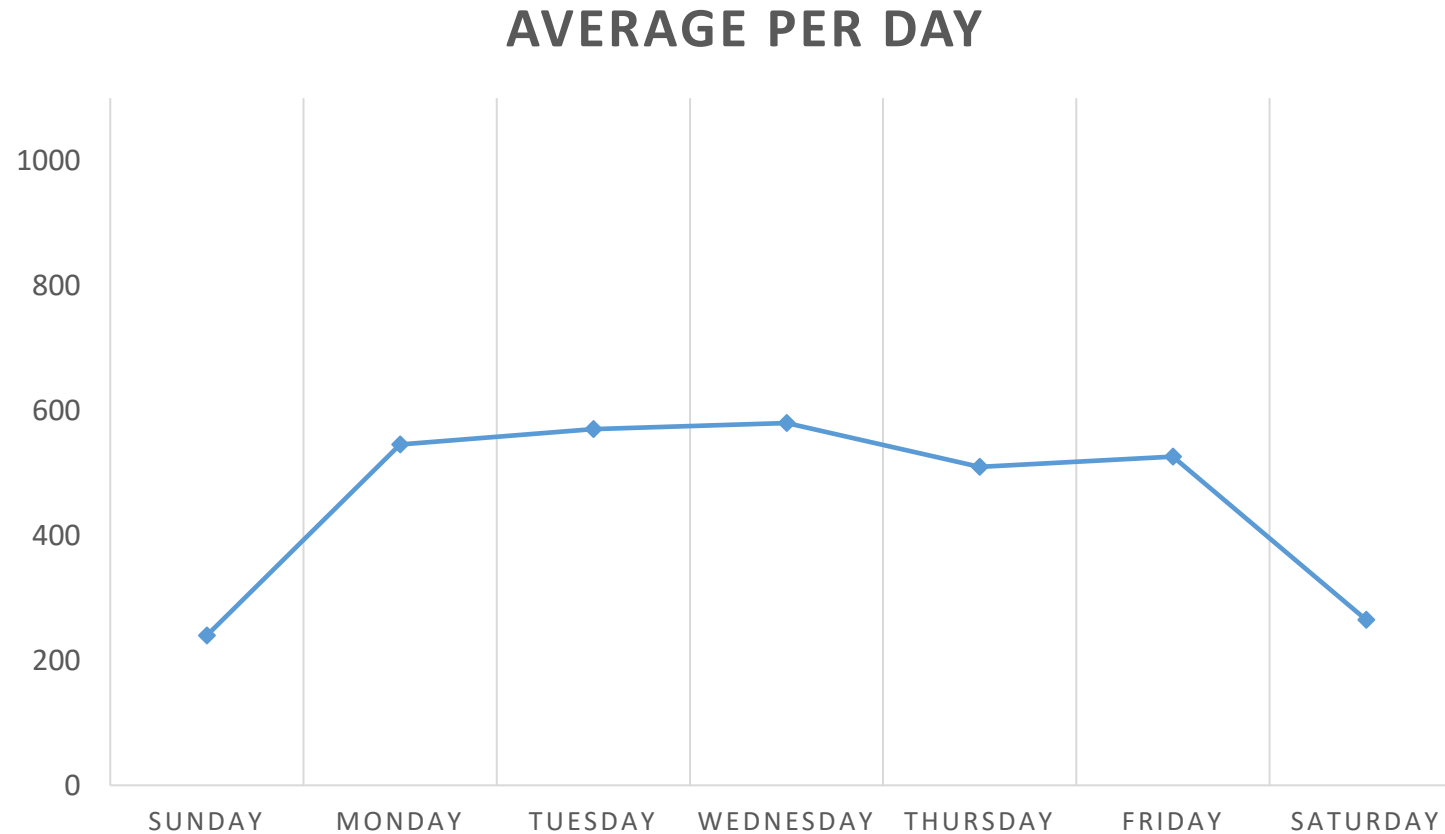
How many dimensions can you identify?



Cycle Plot – Days of the Week mapped over 10 weeks

Complexity

How many dimensions can you identify?



Cycle Plot – Days of the Week mapped over 10 weeks

Complexity

- As visualizations become more complex:
 - They are more challenging to design well
 - Can be more difficult to learn from
- For that reason, visualizations with no more than 3 or 4 dimensions of data are the most common
 - Visualizations with six, seven, or more dimensions can be found but may be more difficult to read!

Complexity

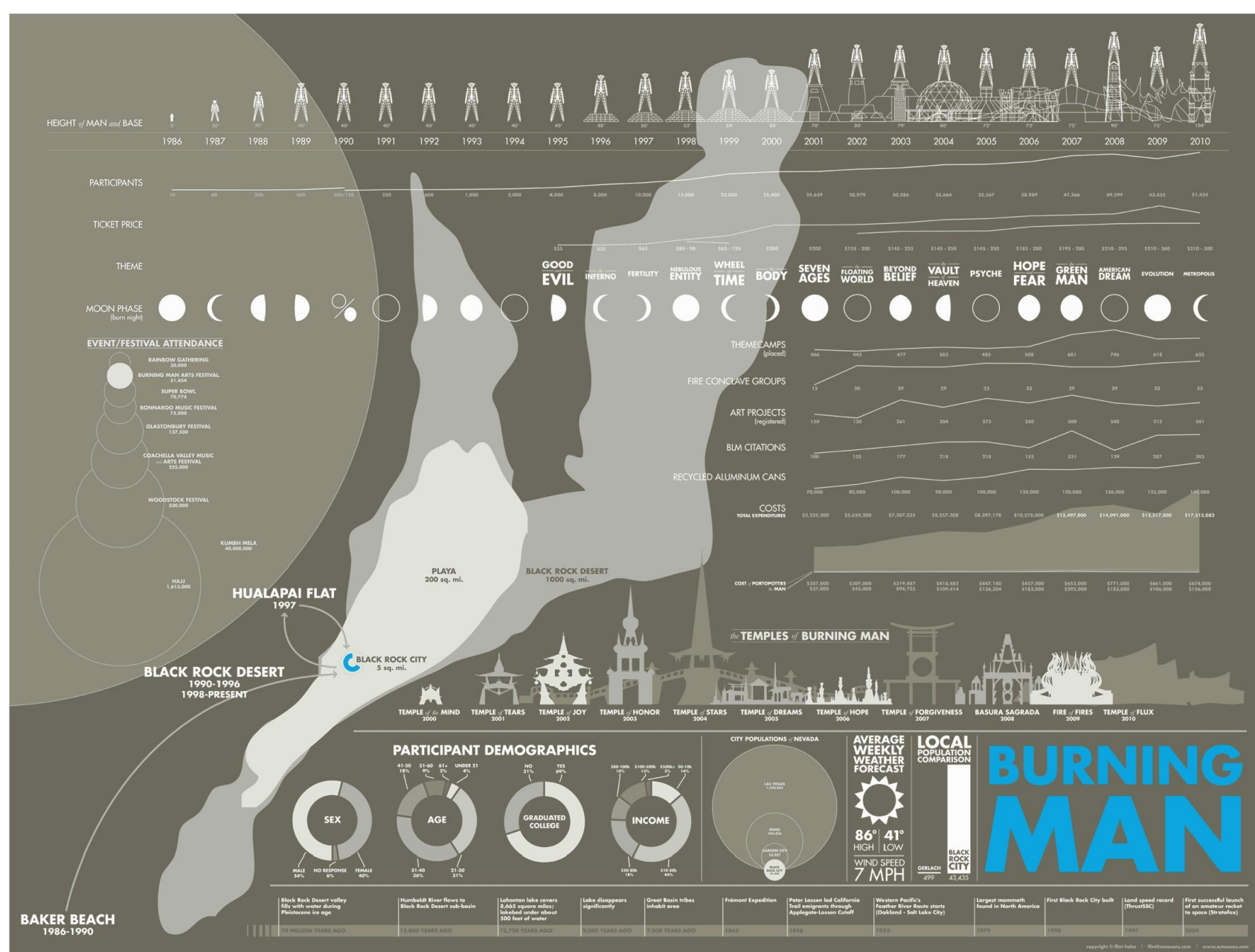
- 2 main challenges to designing more complex visualizations:
 - The more dimensions you need to encode visually, the more individual visual properties you need to use
 - There are relatively few well-known conventions, metaphors, defaults, and best practices to rely on - There is more of a burden on the designer to make good choices that can be easily understood by the reader
 - However, there are smart ways to think about and represent the data

Infographics Vs. Data Visualisation

- **Infographics** and **data visualisation** are slightly nebulous terms often used interchangeably, but a distinction is worthwhile
 - **Infographic** used to refer to representations of information perceived as casual, funny, frivolous. Custom data representation difficult to reuse.
 - **Data Visualization** used to refer to designs perceived to be more serious, rigorous, academic or scientific

Infographics

- Iliinsky & Steele suggest that the term **infographics** is useful for referring to visual representation of data that is:
 - **Manually drawn** (and therefore a custom treatment of the information)
 - **Specific to the data at hand** (and therefore non- trivial to recreate with different data)
 - **Aesthetically rich** (strong visual content meant to draw the eye and hold interest)
 - **Relatively data—poor** (because each piece of information must be manually encoded)



Burning Man, Flint Hahn (2010)

United States 9,400 total warheads

Heading downwards

World's most powerful and reliable arsenal. Some 200 warheads deployed in Europe; additional 2,500

warheads are spares and not counted as operational. Years of balanced reductions with the Russians in both strategic and tactical

warheads. Committed under President Obama to new talks to limit each side to 1,000 strategic warheads



500 tactical

2,126 strategic

France 300 total warheads



300 strategic

Stable A recently renewed arsenal. Small reduction announced in 2008 to less than 300 warheads, but no pressing need for change or updates. Strong political attachment to independent deterrent

China 240 total warheads



180 strategic

Stable Strong commitment to enough weapons to deter attack on its heartland, but no attempt at parity with the US or Russia. Many "strategic" weapons for regional use

UK 185 total warheads



160 strategic

Stable Despite talk of including small arsenal in multilateral talks, no firm commitment to do so. One submarine probably will be cut, but no guaranteed change in total of 160-185 warheads

Israel 200 total warheads



200 strategic

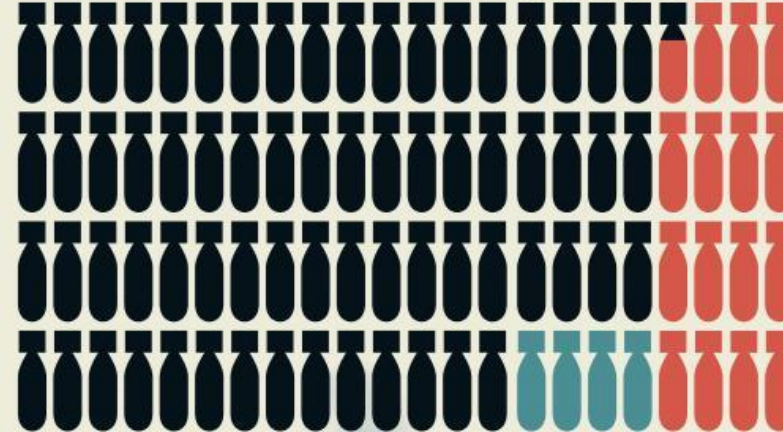
Stable Nuclear weapons never admitted and no details published, but arsenals thought to be about double those of India and Pakistan. A last-resort weapon against Arab invasion

Russia 13,000 total warheads

Heading downwards

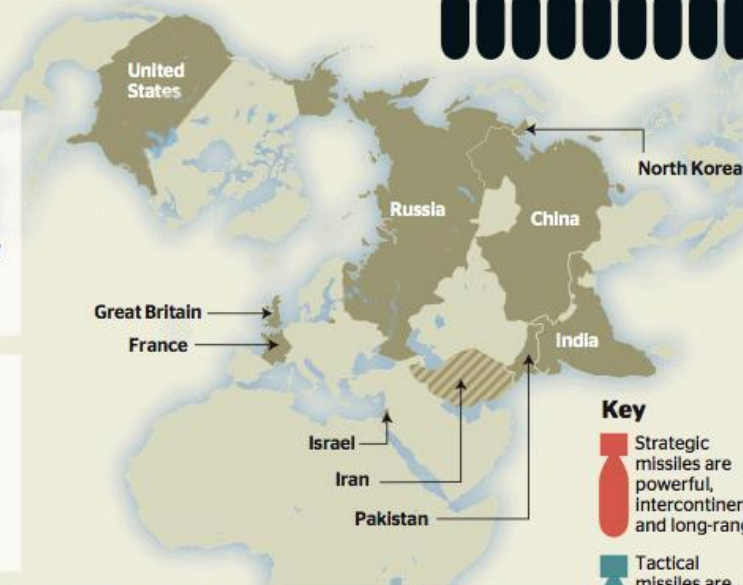
Big reduction from 1991 total of 15,000 warheads. A quarter of present total probably awaiting

dismantling. An ageing arsenal, expensive to maintain and update. Every incentive for further deep cuts



2,050 tactical

2,668 strategic



Key

Strategic missiles are powerful, intercontinental and long-range

Tactical missiles are short-range and battlefield

Weapons in store or awaiting dismantling

Source: FAS

Pakistan 70-90 total warheads



60 strategic

India 60-80 total warheads



60 strategic

Slowly Increasing Approximate equilibrium and no new nuclear arms race likely. All details and number are secret. Only strategic, not tactical missiles, held by each side

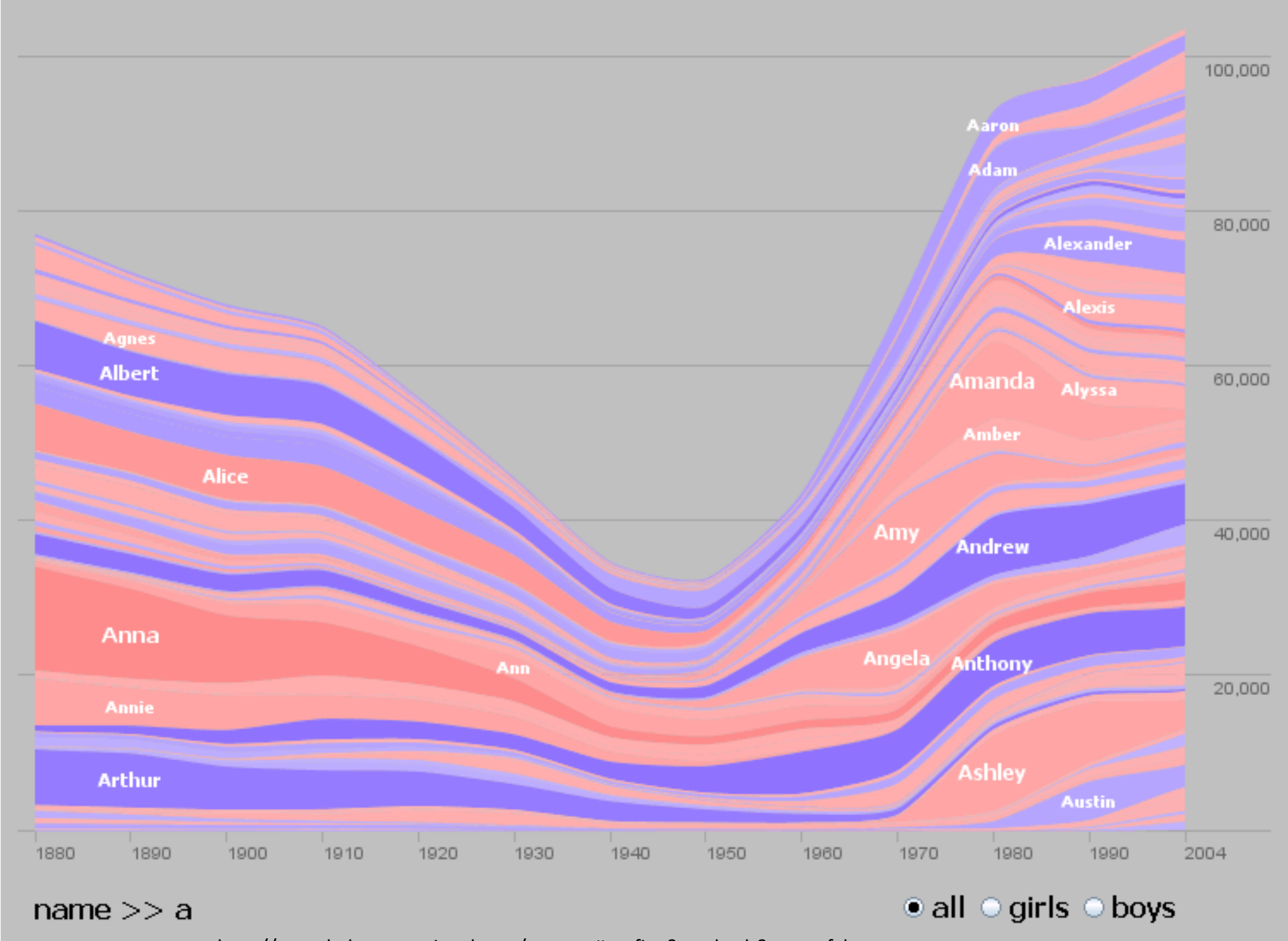
North Korea Fewer than 10 total warheads

Fewer than 10 strategic

Political dependence on hugely expensive nuclear capability. Wants to increase but probably has no more than one or two warheads operational. Intentions depend on present six-power talks to scrap nuclear weapons programme

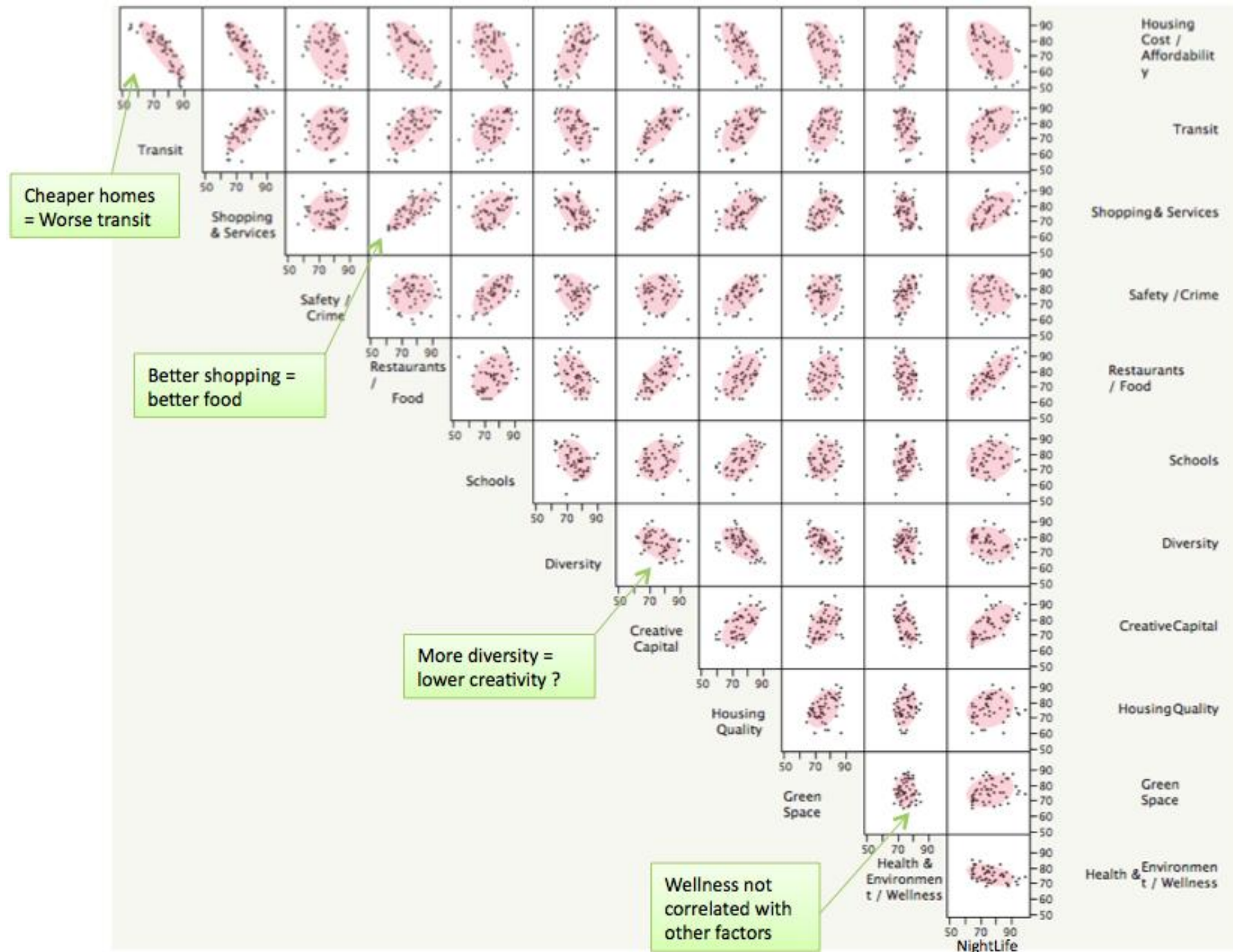
Data Visualization

- By contrast, the term data visualization is useful for referring to any visual representation of data that is:
 - **Algorithmically drawn** (may have custom touches but is largely rendered with the help of computerized methods)
 - **Easy to regenerate with different data** (the same form may be repurposed to represent different datasets with similar dimensions or characteristics)
 - **Maybe aesthetically barren** (data is not decorated)
 - **Relatively data-rich** (large volumes of data are welcome and viable, in contrast to infographics)



Exploratory vs. Explanatory

- Exploratory data visualizations are appropriate when you have lots of data and you are not sure what's in it
 - - When you need to get a sense of what's inside your data set, translating it into a visual medium can help you quickly identify its features, including interesting curves, lines, trends, or anomalous outliers
- This type of visualization is typically part of the data analysis phase, and is used to find the story the data has to tell you



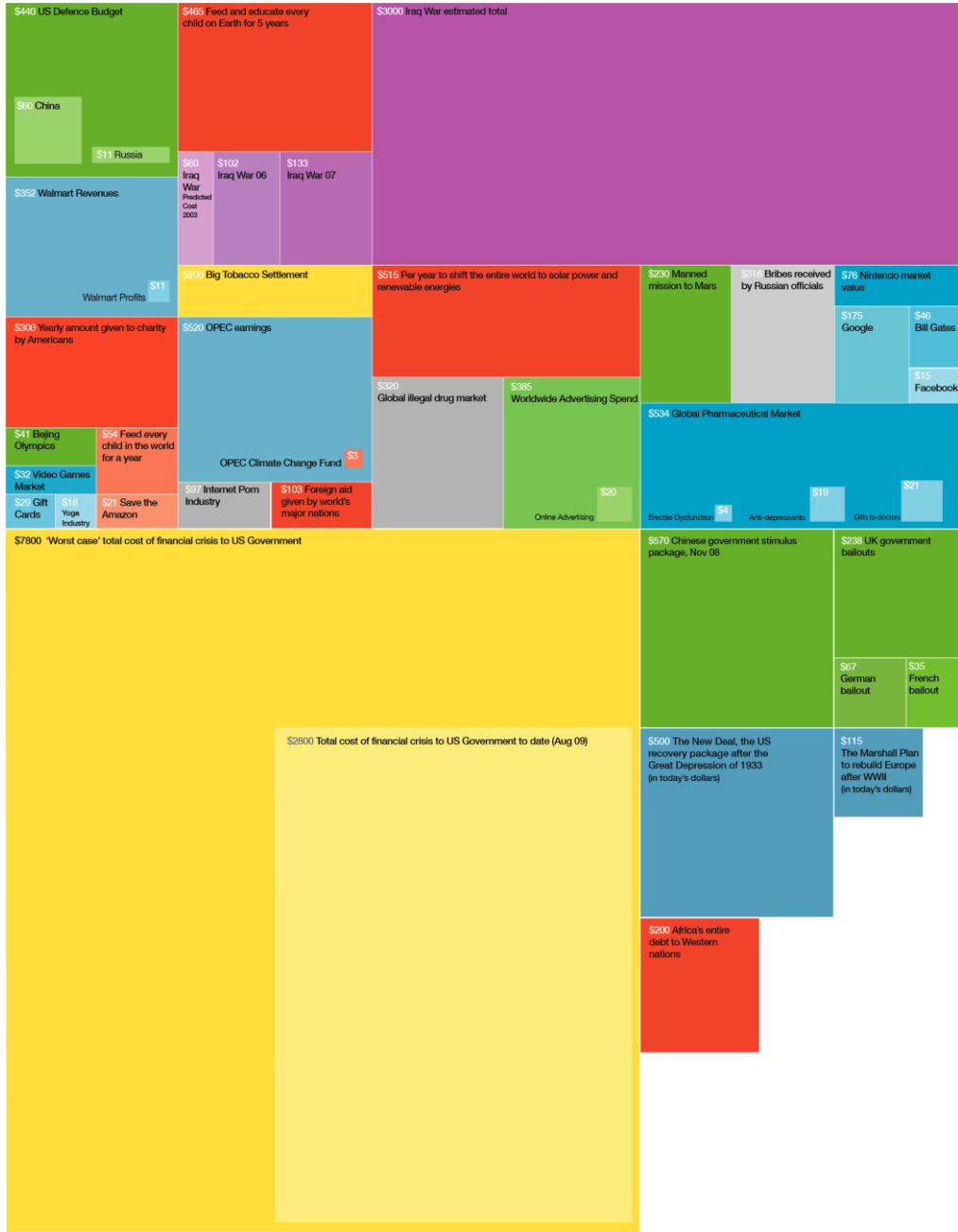
Exploratory vs. Explanatory

- Explanatory data visualization is appropriate when you already know what the data has to say, and you are trying to tell that story to somebody else
 - Whoever your audience is, the story you are trying to tell is known to you at the outset, and therefore you can design to specifically accommodate and highlight that story
- Explanatory data visualization is essentially part of a presentation phase

The Billion Dollar Gram

Billions spent on this. Billions spent on that. It's all relative right?

■ spending ■ earning ■ giving ■ fighting ■ losing ■ illin'



source: New York Times, The Guardian, Fortune and others // see: informationisbeautiful.net for full list

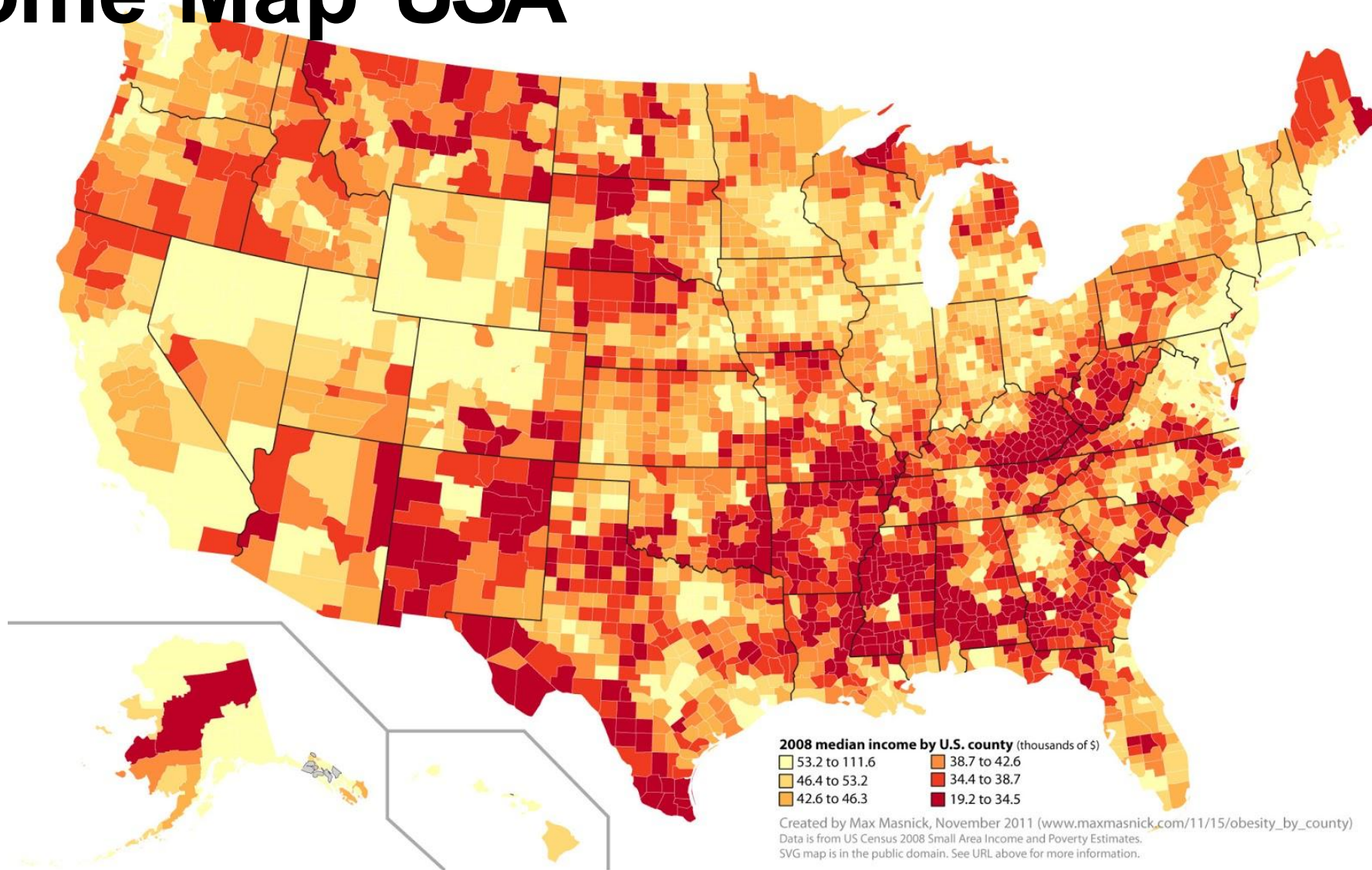
David McCandless
InformationIsBeautiful.net

<https://www.informationisbeautiful.net/visualizations/the-billion-dollar-gram/>

Hybrids: Exploratory & Explanatory

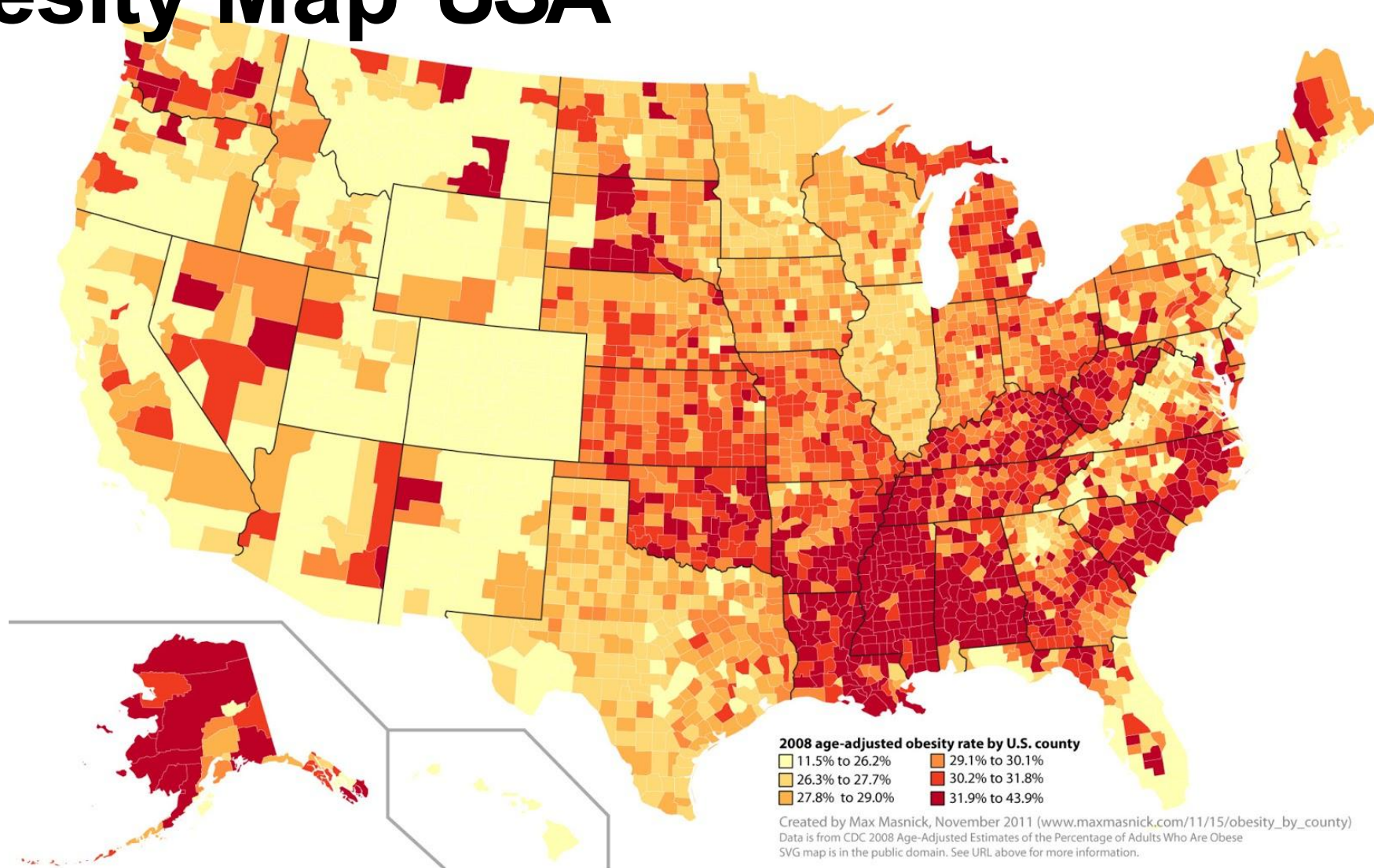
- Hybrid category: a curated dataset that is nonetheless presented with the intention to allow some exploration on the reader's part
 - These visualizations are usually interactive via some kind of graphical interface that lets the reader choose and constrain certain parameters
 - Sometimes allows the reader discover insights the designer of the visualization hasn't come across yet!
- In these hybrid designs there is a certain freedom- of-discovery aspect to the information presented, but it is usually not totally raw; it has been distilled and facilitated in some way

Income Map USA



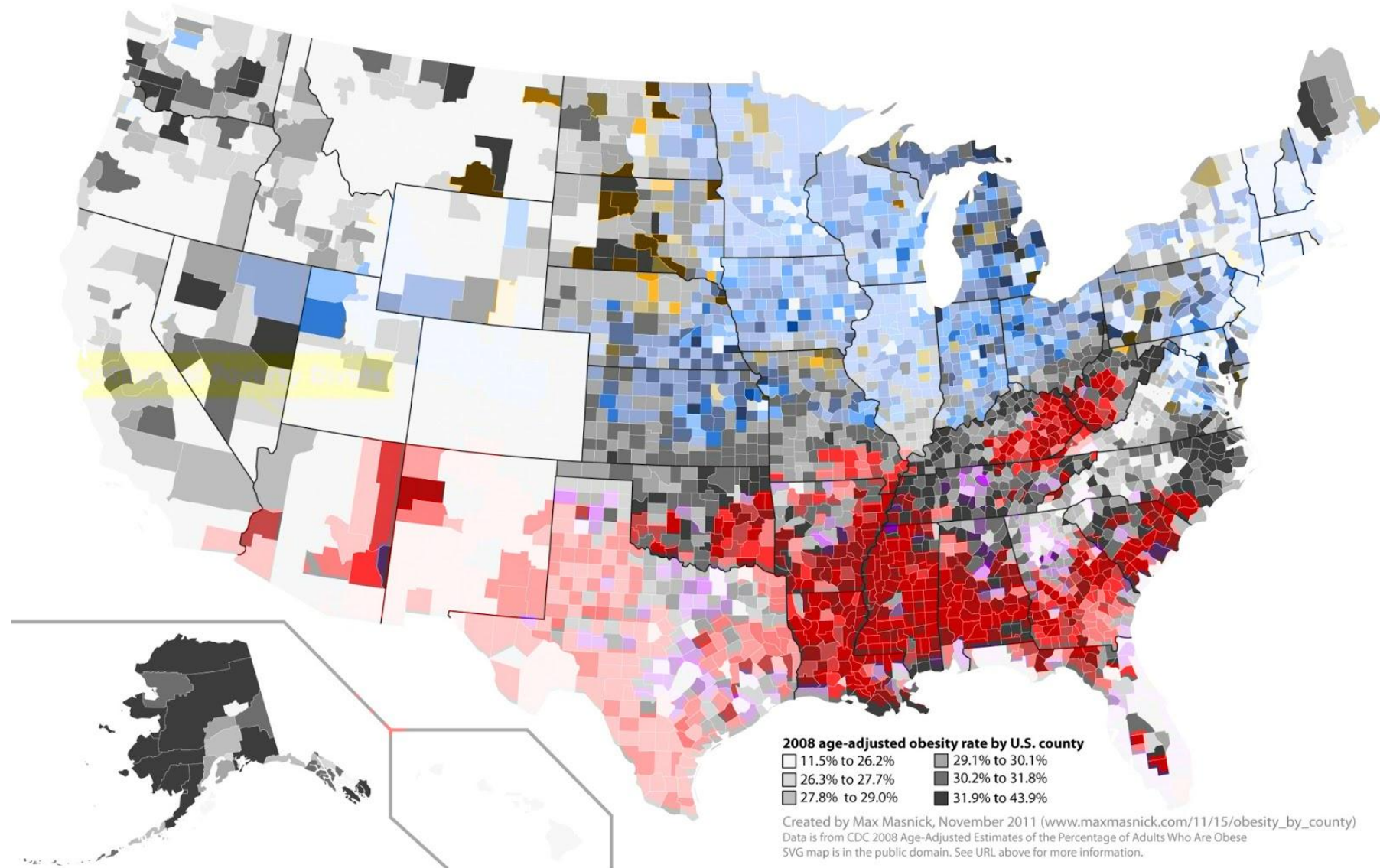
<http://asserttrue.blogspot.ie/2013/02/poverty-and-obesity-in-america-how-they.html>

Obesity Map USA



<http://asserttrue.blogspot.ie/2013/02/poverty-and-obesity-in-america-how-they.html>

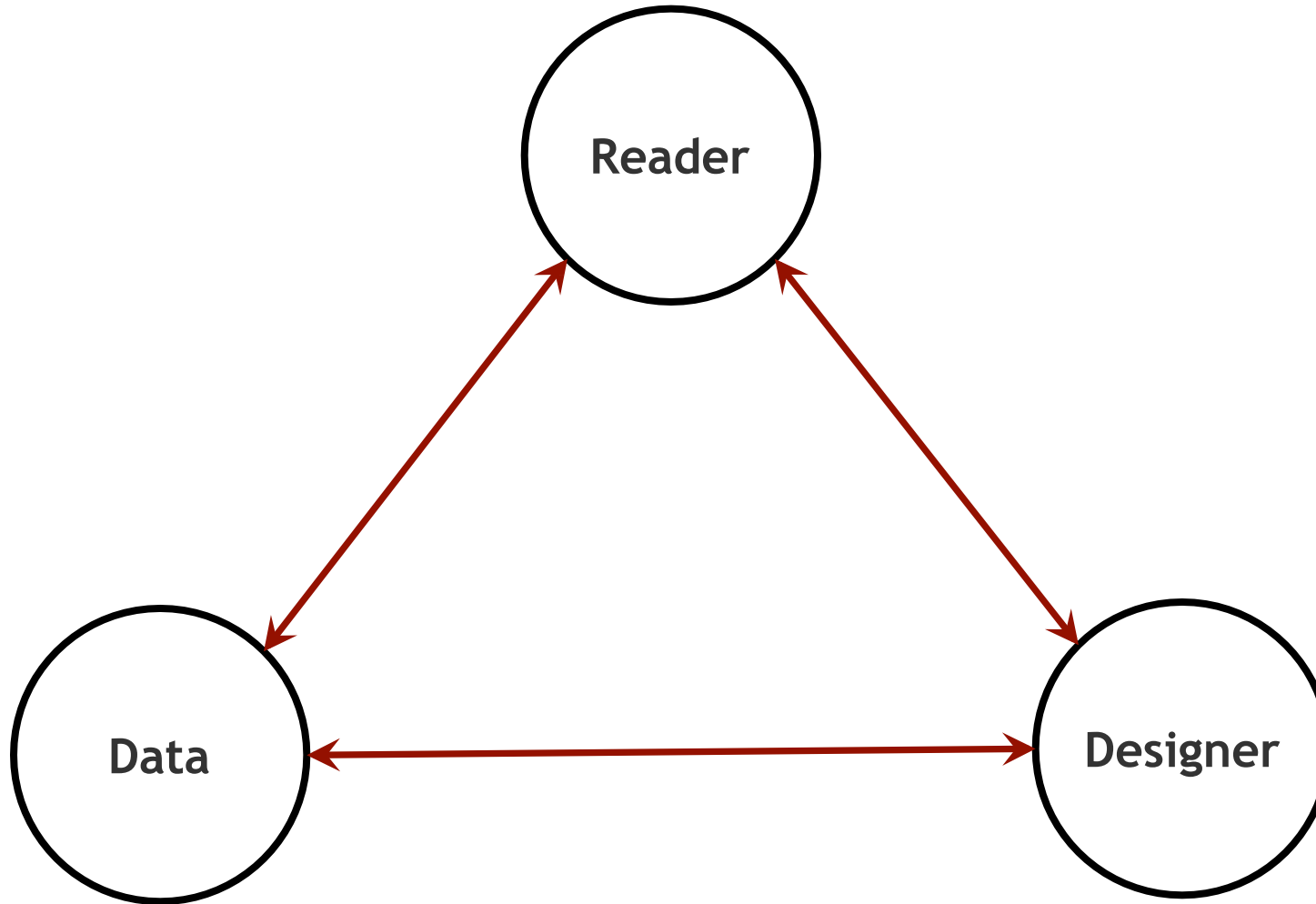
Correlated Map



Areas of dark red are where very high obesity rates and lowest income levels coincide

Informative - Persuasive - Visual Art

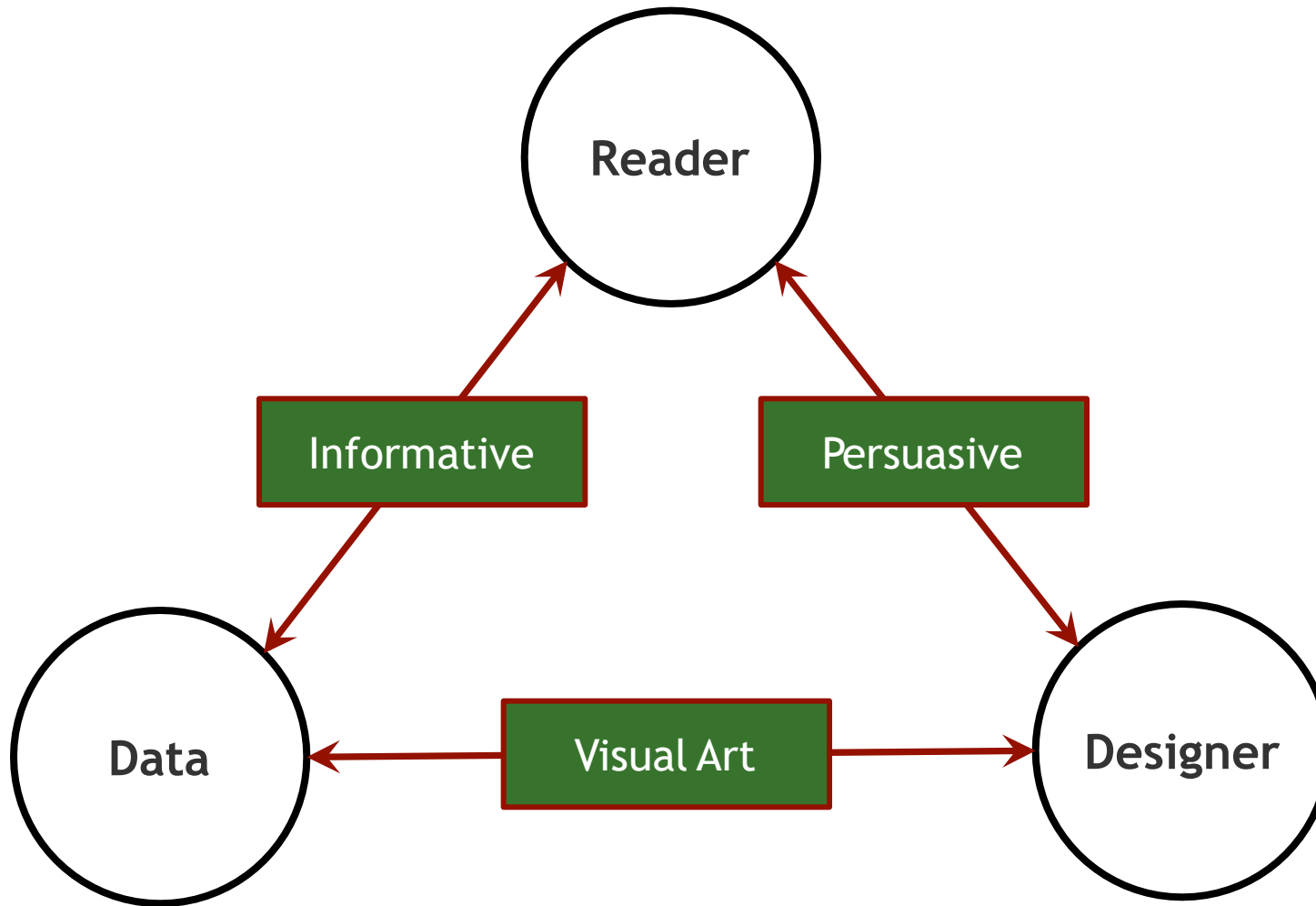
The Designer-Reader-Data Trinity



“Designing Data Visualizations”, N. Iliinsky & J. Steele, O'Reilly Media, 2011

<http://shop.oreilly.com/product/0636920022060.do>

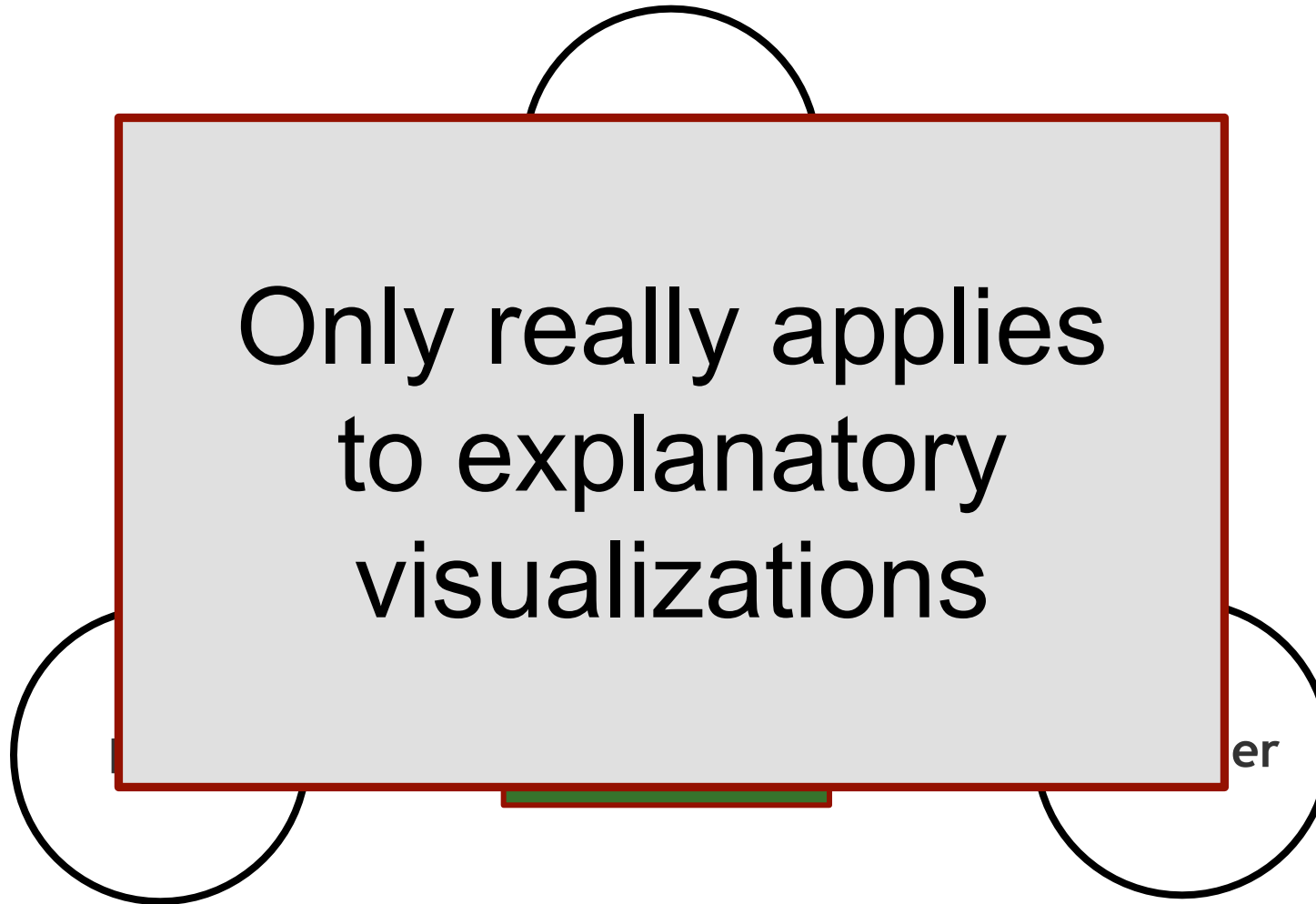
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“Designing Data Visualizations”, N. Iliinsky & J. Steele, O'Reilly Media, 2011

<http://shop.oreilly.com/product/0636920022060.do>

The Designer-Reader-Data Trinity

- We can think of explanatory data visualization as being supported by a three-legged stool consisting of the designer, the reader, and the data
- Each of these "legs" exerts a force, or contributes a separate perspective, that must be taken into consideration for a visualization to be stable and successful
- Each of the three legs of the stool has a unique relationship to the other two
- While it is necessary to account for the needs and perspective of all three in each visualization project, the dominant relationship will ultimately determine which category of visualization is needed

Informative

- An informative visualization primarily serves the relationship between the reader and the data.
- It aims for a **neutral presentation** of the facts in such a way that will educate the reader (though not necessarily persuade them).
- Informative visualizations are often associated with broad data sets, and seek to distil the content into a manageably consumable form.

Gamestop's share price



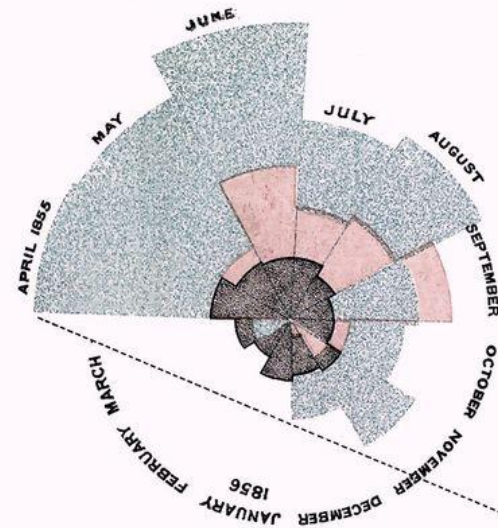
Source: Google search Gamestop share price

Persuasive

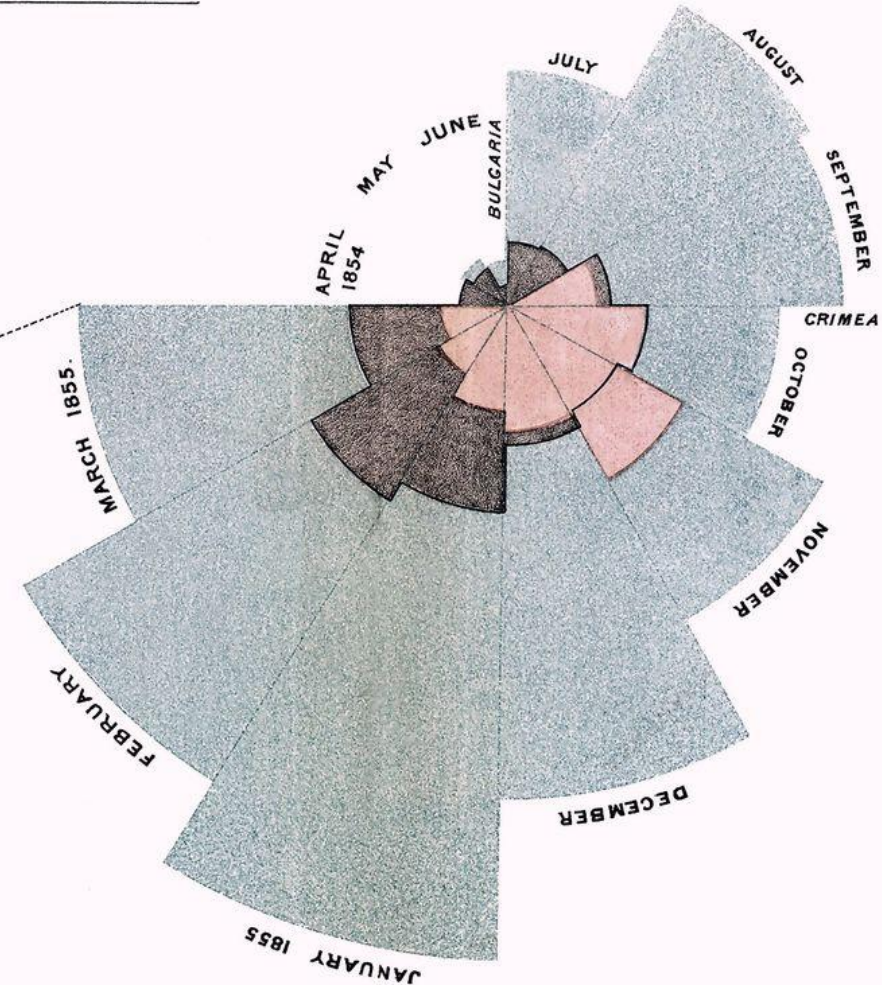
- A persuasive visualization primarily serves the relationship between the designer and the reader.
- It is useful when the designer wishes to change the reader's mind about something.
- In this category of visualization, the data represented is specifically chosen for the purpose of supporting the designer's point of view, and is presented carefully so as to convince the reader of same.

DIAGRAM OF THE CAUSES OF MORTALITY IN THE ARMY IN THE EAST.

2.
APRIL 1855 TO MARCH 1856.



1.
APRIL 1854 TO MARCH 1855.



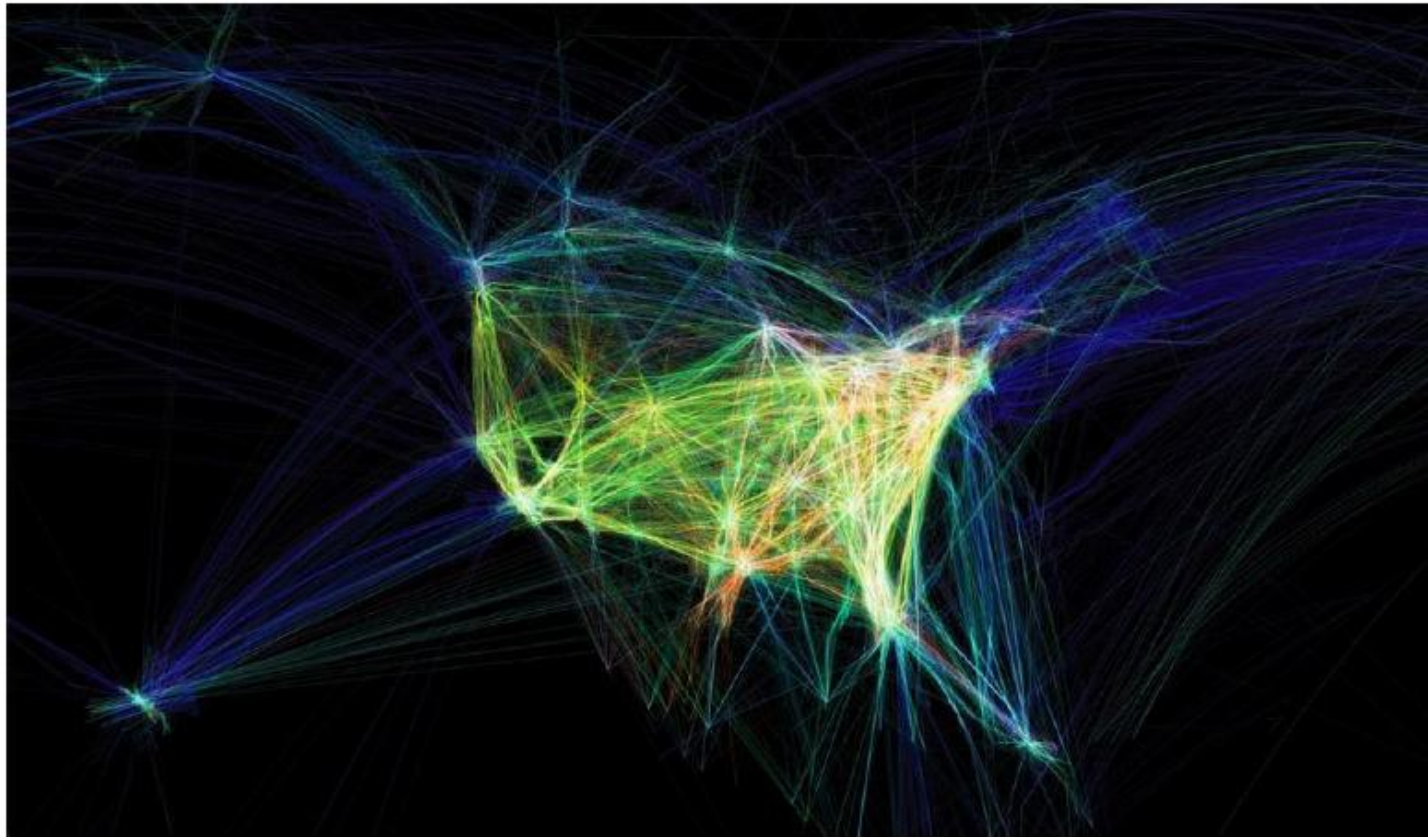
The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex.
The blue wedges measured from the centre of the circle represent area for area the deaths from Preventable or Mitigable Zymotic diseases; the red wedges measured from the centre the deaths from wounds; & the black wedges measured from the centre the deaths from all other causes.
The black line across the red triangle in Nov^r 1854 marks the boundary of the deaths from all other causes during the month.
In October 1854, & April 1855, the black area coincides with the red; in January & February 1856, the blue coincides with the black.
The entire areas may be compared by following the blue, the red & the black lines enclosing them.

Visual Art

- The third category, visual art, primarily serves the relationship between the designer and the data.
- Visual art is unlike the previous two categories in that it often entails **unidirectional encoding of information, meaning that the reader may not be able to decode the visual presentation to understand the underlying information.**
- The designer may intend only to condense it, translate it into a new medium, or make it beautiful; she may not intend for the reader to be able to extract anything from it other than enjoyment.

Visual Art

Flight patterns Aaron Koblin



<https://www.artic.edu/artworks/213139/flight-patterns>

FaceBook Visualised



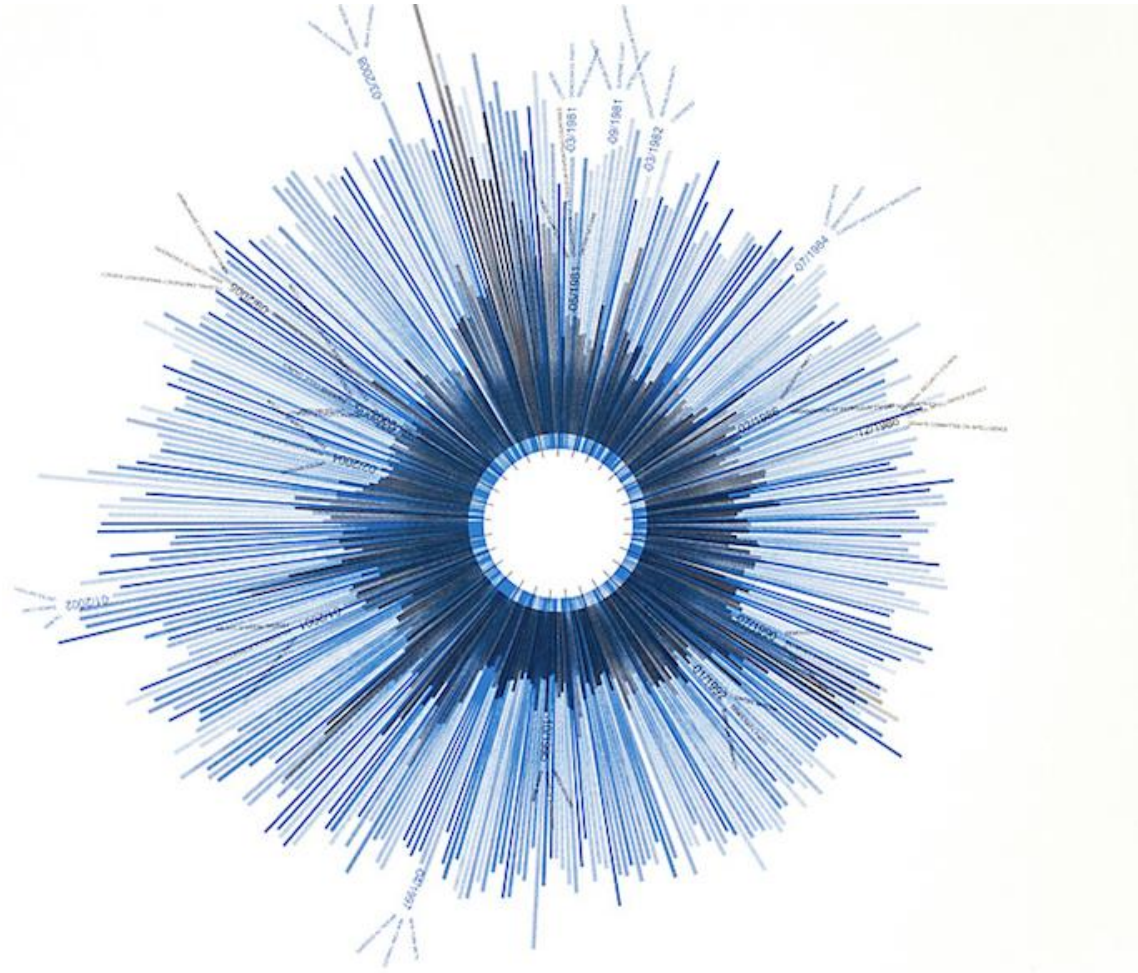
<http://edition.cnn.com/2010/TECH/social.media/12/14/facebook.relationships.mashable/index.html>

How often the NY Times printed the words “hope” and “crisis” between 1981 and 2010 by Jer Thorp

Each bar represents a month

Dates and mentions of specific events and key words are thrown in here and there to orient the viewer.

It's interesting to note those times when crises eclipsed hope.



Classifications of Visualisations

- There are many different ways to classify visualisations
- Classifications can be useful when discussing/planning visualisation projects
- Classifications are also useful when determining the appropriate visualisation techniques to deploy

Interesting reading

- Open Mind Blog.
- **USA Temperature: can I sucker you?**
- Link: <https://tamino.wordpress.com/2018/08/08/usa-temperature-can-i-sucker-you/>

Video

Weather visualisation

https://youtu.be/q01vSb_B1o0?t=46

Thanks To

Cathy Ennis, John McAuley, Colman McMahon and
Brian Mac Namee
for an earlier version of these lecture notes

Short Glossary of Terms

- **Reader:** The consumer of a visualization, often someone other than the designer.
 - The reader has information needs that are meant to be satisfied by the visualization.
- **Infographic:** Visualizations that are manually generated around specific data, tend to be data-shallow, and are often aesthetically rich.

Short Glossary of Terms

- **Data Dimension:** One single channel of data (e.g. a stock graph may comprise dates and prices - each is a unique dimension)
- **Variability of a data dimension:** The values that are present or allowed in a single data dimension (e.g. integers vary discretely; position can vary continuously; categories are finite and discrete, though may be hierarchical; numbers are infinite)

Short Glossary of Terms

- **Visual Property:** A characteristic that you can see (e.g. colour, size, location, thickness, and line weight)
- **Encoding:** The visual property applied to a dimension of data that encodes the information into a visual medium for decoding by the reader.